

The Verticality Effect: Exploring the Relationship Between Vertical Space Navigation and Player Presence in Modern 3D Digital Games

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Abstract

This thesis investigates how verticality in modern 3D digital games functions as a key design element shaping player experience and fostering presence. While spatiality and player presence are well-established topics in game studies, the specific experiential impact of vertical space navigation remains largely underexplored. This study addresses this gap by establishing a multidisciplinary theoretical framework that synthesises theories of player involvement, presence and presence. Through an interpretive examination of the theories and qualitative analysis of case studies, the research demonstrates how verticality operates in relation to player experience.

The findings reveal that verticality significantly enhances player engagement by expanding player agency, strengthening kinesthetic involvement through embodiment and acting as a medium for environmental storytelling. The research concludes that by simultaneously activating different forms of involvement, verticality serves as a valuable vector for presence, making virtual worlds more tangible, meaningful, and experientially rich. This study also provides a robust framework for analysing verticality, offering valuable insights for both game scholars and designers seeking to create more compelling experiences in digital games.

Keywords:

Verticality, presence, experience, space, games

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Chapter 1: Introduction

When asked why he sought to climb Mount Everest, the British mountaineer George Mallory famously replied, “because it’s there”. Since then, the phrase has echoed as a symbol of human curiosity and the drive to reach beyond one’s limits. But behind its simplicity lies a profound truth: climbing, ascending, and standing on lofty heights has long stirred the human imagination. The higher we rise, the more heightened our emotions seem to get. Similarly, the climber Alex Honnold, known for his ropeless ascents of steep rock faces, is also a figure who often talks about the feeling of pure exposure, the incredible views, the sense of being totally present that arises from a climb (Honnold, 2025). Such experiences are not limited to climbing or mountaineering. Even casual, everyday acts like going to a rooftop café, admiring a city landscape from the window, or gliding down a hillside on a bike has a considerable impact on our senses. Height has a powerful influence on the deep human desire that is to relate to the world. This blend of heightened perspective, human challenge, and often, a sense of being more present is a way to feel more alive. A quiet thrill runs through the body as we stand high above the world, the gaze expands toward the horizon while the ground falls away beneath us. The air feels thinner, the landscape stretches wider, and our thoughts often become sharper, more focused. We enjoy the view, concentrate on our next step upwards, adrenaline pumps in the body when we look down. This physical sensation of exposure evokes a unique feeling of vulnerability that mixes with empowerment.

Digital games, too, provide ways to experience the exhilaration of verticality. With the evolution of technology and design, virtual environments are able to feature impressive capabilities in three dimensions (3D). Players can now climb towers, jump across rooftops, slide down into caverns, and glide toward a far-reaching goal with a dimensional perspective that is closer to the player. These vertical interactions, while adding movements, also shape how players perceive, navigate, and inhabit the game world. They evoke the same sensations of awe and tension that real-world verticality inspires, or do they?

In this thesis, I explore the significance of verticality in game design. I argue that vertical space in digital games and the navigational possibilities it holds are not neutral features of navigation, but rather meaningful components that heighten the presence of a player in a digital environment, a feeling of “being there” in a virtual world. The way players ascend, descend, and move through vertical structures contributes to how they feel situated, embodied, and emotionally connected to the

environment. As such, verticality holds serious potential for deepening our understanding of player engagement and presence in digital games.

The concept of presence is a key aspect of understanding player experience in digital games. Finding its origins in fields such as virtual reality, human-computer interaction, and cognitive psychology, presence in game studies is often discussed as a measure of how convincingly a digital space can simulate a feeling of “being there” (Lombard & Ditton, 1997; Slater & Wilbur, 1997). Calleja’s (2011) model of player involvement provides a comprehensive framework for understanding player presence in digital games. He identifies six dimensions of player involvement that act as the building blocks that collectively produce the sense of “being there” within a game world. In this view, presence is an emergent quality shaped by how players engage with a game and to which level.

At the same time, space in digital games is a designed, navigable, and interpretable environment. Game worlds are spatial constructs in which players perform and experience liveable moments. Scholars like Yi-Fu Tuan (2001), Casey (2002) and Henri Lefebvre (1991) have extensively argued that space is not neutral, but rather lived, produced through culture, and constructed with meaning that emerges through action and memorable moments. These notions of space have been extensively drawn on in game studies, claiming that the experience of space in digital games stimulates relatable sensations (Calleja, 2011; Murray, 2017; Nitsche, 2008).

While both presence and space have received considerable attention in game studies, the specific phenomenon of verticality remains underexplored. Most research on the subject presents the experience of verticality as part of the global instance of space and player engagement. However, in order to have applicable and clear design tools to produce digital games that contribute to positive player experiences, it is worth diving deeper and analysing the effect of verticality on player experience in a more resolute and thorough manner. It is a critical aspect that offers an innovative direction for rethinking the relationship between space and player presence. This thesis emerges from the intersection of these three core concepts: presence, space, and verticality yielding the following research question.

How does verticality influence player experience and presence in modern 3D digital games?

I specify 3D in this research question because I recognise that verticality is a phenomenon that takes on a qualitatively different role in three-dimensional game worlds compared to two-dimensional ones. While early 2D platformers such as *Donkey Kong* (Nintendo, 1981) or *Ice Climber* (Nintendo, 1985) introduced basic vertical mechanics like jumping or climbing a ladder, their composition remained

confined to a flat, side-scrolling plane. In such games, verticality functions primarily linearly to present obstacles or challenges. Indeed, players ascend or descend along a predetermined path, and vertical movement typically lacks depth. In contrast, 3D games provide a far more nuanced and rich engagement with vertical space. Verticality allows players to perceive and navigate in depth and elevation, it creates volume. It demands greater awareness of the space because the vertical structures surround the player and challenges their orientation. Furthermore, and as it will be discussed in this research, 3D spatial design allows verticality to offer much more symbolic meanings that are harder to convey in 2D games where the lack of camera control, volumetric space, and embodied navigation limit the expressive potential of vertical structures. Focusing on modern 3D digital games therefore provides a richer foundation that does not only depart from a linear verticality. This way, I am able to analyse how verticality shapes presence across multiple dimensions.

To answer my research question, the study adopts a theoretical, interpretive methodology that builds upon existing literature in game studies, spatial theory, and presence research, and considers how verticality may impact different types of player involvement. The objective is to clarify the design affordances of verticality across multiple experiential dimensions identified by Calleja's (2011) involvement model such as how it affects movement and player agency, how it structures the raw feeling of a game through inputs, and how it supports narrative construction. The study also aims to ground these theoretical claims in detailed case studies that illustrate how vertical space shapes gameplay and presence in contemporary digital games. As this research is theoretical and analytical, it draws on foundational literature in game studies, particularly Gordon Calleja's (2011) work on player involvement and immersion and combines game studies texts with theorists of space such as Henri Lefebvre (1991), Yi-Fu Tuan (2001), and Edward Casey (2002). These authors' concepts of lived space, place-making, and spatial meaning are applied to digital games and linked to Gibson's (2015) ecological approach of perception which offers a strong basis for understanding game affordances. In terms of player presence, the thesis draws from the original ideas of Minsky (1980) on telepresence and expands on the further research realised on the concept from scholars like Lombard & Jones (2015), Wirth et al. (2007) and Witmer & Singer (1998). In addition to these theoretical frameworks, the analysis applies selected case studies of modern 3D digital games that exemplify verticality and provide practical demonstration of its impact on player presence. These include *The Legend of Zelda: Tears of the Kingdom* (Nintendo, 2023), *The Finals* (Embark Studios, 2023), and *Fallout 4* (Bethesda, 2015). Each game is examined through the specific aspects of verticality I identified: game affordances, kinesthetic involvement and narrative. These analyses aim to demonstrate how verticality operates within the structure of gameplay and spatial design, and how it affects player experience by impacting player engagement. With this methodology, this research contributes to broader conversations in

game design regarding player experience and player presence as well as how verticality can be used as an impactful design tool.

This thesis is structured in five chapters including this introduction. Each chapter builds toward the central argument that verticality plays a significant role in shaping player presence in modern 3D digital games by engaging multiple dimensions of player involvement. Chapter 2 reviews the literature around the theories of presence from cited scholars and presents the conceptual foundations for understanding player presence in games. It introduces Gordon Calleja's (2011) model of player involvement as a framework for analysing how players reach presence through the simultaneous participation of the different dimensions of player engagement on which verticality will be analysed. As a supporting theoretical framework, chapter 3 presents the theories around the human experience of space and explores how space in digital games is perceived, constructed, and felt. The chapter also discusses theories of perception and emphasises Gibson's (2015) affordance-based notion of perception. This chapter establishes key principles of space that will be revisited through the perspective of verticality. Chapter 4 is about verticality in game design and is the core analytical chapter. It defines verticality and explores its dimensions that influence player involvement. The chapter is divided into four sections: verticality as affordance, verticality and game feel, verticality as a narrative device, and a final synthesis that positions verticality as a significant factor in enhancing player presence. Here, the multiple case studies of modern 3D digital games are used to ground the discussion in practical demonstrations. Chapter 5 is the concluding chapter. It synthesises the findings and reflects on the significance of verticality as an engaging feature in digital game design. It highlights how verticality stimulates various types of player involvement and contributes to a meaningful sense of presence. Here, I also consider the broader implications of verticality, suggesting future research directions for designing valuable verticality for digital games.

This research contributes to the ongoing discourse in game studies by offering a focused and in-depth investigation of verticality as a design feature that significantly shapes the player's experience of presence in modern 3D digital games. While verticality has never been an aspect of game design that has been studied in depth, this thesis argues for its central role. Moreover, the significance of this research lies in its interdisciplinary approach. Drawing from game studies, spatial theory, environmental psychology, and phenomenology, this study integrates diverse perspectives to develop a richer understanding of how players interact with verticality. The use of the player involvement model (Calleja, 2011) as a conceptual framework allows for a structured analysis of how vertical design influences forms of engagement consequently acting as a building block for player presence. In a broader cultural and psychological context, this research touches on fundamental questions about

why spatial experience matters. Humans have long been drawn to vertical spaces like climbing mountains, building towers, positioning temples above a long set of stairs, or simply seeking higher ground for a better view. This vertical impulse challenges physical experiences but also conveys symbolic meanings that are familiar to human societies. By exploring how these impulses are stimulated and transformed within digital game environments, this thesis can reveal the expressive potential of verticality to mirror, extend, or simply understand our real-world spatial experiences. Lastly, this study is well-timed. As contemporary 3D games grow increasingly complex in their design, understanding the impact of verticality becomes crucial for both designers and scholars. The findings of this research may inform future design practices, helping developers consider verticality as an expressive medium useful for creating engaging artefacts. Let us now dive into this compelling research and explore this thesis' first foundation: the notion of player presence.

Chapter 2: Presence in Virtual Environments

2.1 Introduction

This chapter explores the concept of presence in virtual environments, with a particular focus on its role in player experience in digital games. It draws from theories coming from multiple disciplines in order to define the concept of presence and distinguish it from related ideas like immersion. It is an essential theoretical foundation to support later discussions on how game design and, particularly the use of vertical space, can intensify or modify a player's experience.

Let us imagine a prisoner in Plato's Allegory of the Cave, chained in darkness, perceiving only shadows on the wall. To them, these projections are reality, until one day, they are freed and step into the light, realising the world they knew was merely an illusion. This ancient parable reflects a fundamental question about human perception and reality: What makes an experience feel real? In the modern era, virtual reality (VR) poses a similar question. When users enter a VR environment, they are not physically transported to another world, yet their senses and cognition may convince them otherwise. The shadows on the cave wall have evolved into high-resolution headsets, spatial audio, and motion tracking, creating illusions that try to blur the boundary between the digital and the real. Like the freed prisoner experiencing the world for the first time, players in virtual environments encounter constructed realities that feel tangible, even though they are simulations. This sensation of "being" in another place is at the heart of what researchers call presence (Calleja, 2011; Lombard and Jones, 2015; Minsky, 1980; Slater, 2003).

In the film *Ready Player One* (Spielberg, 2018), the audience is transported into the OASIS, a vast and immersive virtual reality universe where players can escape their everyday lives and exist in a fully realised digital world. The protagonist, Wade Watts, finds comfort, adventure, and belonging within this world. The film portrays a future where this virtual world is so compelling that the boundaries between physical and virtual reality blur.

Even though this is portrayed in science fiction, virtual reality we know today comes with the promise of making such experiences part of our lives. It seems like a human dream to be able to enjoy an all-encompassing second reality that allows us to be in a different place than the one we are in at the moment (Grau, 2003). Literature and cinematography tell us that, but also history. The idea that different forms of media, art and technology sought to simulate other worlds isn't exclusive to modern technology. For instance, Roman villa frescoes (around 60 BCE) were designed to mimic an alternate

reality, an illusion of spaces to transport viewers somewhere else (Grau, 2003). During the Baroque period in the 17th century, artists like Caravaggio and architects like Bernini used dramatic lighting, perspective and dynamic compositions to create immersive experiences, called Baroque illusionism (Meyer, 2022). Bernini's "Ecstasy of Saint Teresa" in Rome has been celebrated for its dramatic emotional expression and dynamic composition, which combines sculpture, trompe l'oeil fresco art and light to bring the public into a narrative (Devinda, 2024; Meyer, 2022). All of these in the attempt to immerse the audience into enjoying their work by helping them get involved with it.

This determined attempt to reduce the perceived gap between the audience and the environment illustrated in the media, which often creates a feeling of "being there", is referred to as *presence* or *immersion*. In digital games, authors suggest that the experience of presence is the result of a player's high level of involvement when interacting with the digital environment (Calleja, 2011; Mcahan, 2003; Witmer & Singer, 1998). Because immersion and presence have been used interchangeably to explain the same kind of experience even though they are not synonyms, it also becomes problematic for researchers to give meaning to the phenomenon.

Immersion has been used to signify various types of human experiences across many different disciplines like media studies, psychology, art and digital games (Agrawal et al., 2020). "Immersive" has been used for distinct meanings. A 3D digital game with realistic graphics, sounds and physics could be claimed to be immersive. Similarly, a more simple virtual art tour, such as Google's virtual exposition portraying Picasso's Guernica was claimed immersive. It was argued that a combination of interactivity, historical context, and engagement immerse the audience into the digital tour (Google Arts & Culture, 2025; Sebiru, 2025). On this basis, Murray (2017) highlights the transformative potential of digital media in storytelling. She discusses how digital environments can create immersive experiences by allowing users to participate actively in narratives, thus reshaping how stories are told and experienced. The author expresses that immersion is a psychological experience that seeks to be similar to the physical experience of being submerged in water. Being surrounded by another reality the same way water surrounds us when we are immersed. However, the discourse around immersion is so widely researched in numerous disciplines that major perspectives arise. Mainly there are two major visions on immersion: either it is determined by technology (Slater, 2003) or by a person's state of mind (Witmer and Singer, 1998). This suggests that immersion is a multidimensional concept where the academic discourse fails at finding a common meaning. Yet, when looking at the concept of presence for clarity on the similar phenomenon, it has equal complexity.

The term presence is derived from *telepresence*, which was coined by Minsky (1980) to describe the sensation of controlling machines from a distance as if they were present at this location while being

physically situated within their operating environment. This sensation is created through a combination of the use of advanced sensors, actuators, and communication systems that replicate the person's motions and sensory experiences in a remote environment. The idea is that machines could be able to perform tasks with similar dexterity and sensitivity as human hands. This encompasses the idea of receiving sensory feedback, so the operator can see, feel, and interact with their surroundings as if they were truly there. However, Minsky outlines several challenges to developing such systems: Current remote-control devices lack the skill and nuance of human movement. Translating physical sensations like touch, temperature, or pressure in a way that the operator can intuitively understand and respond to requires advanced research and significant financial investment. Finally, upgrading or retrofitting current facilities to welcome telepresence systems presents logistical challenges.

Lombard and Jones (2015, p.14), in their chapter "Defining Presence", explain the complex nature of the term presence. The presence phenomenon has multiple interpretations, which resulted in different and divergent definitions of the concept. It has been widely explored in disciplines such as psychology, communication studies and human computer interaction, with scholars attempting to define and measure the phenomenon in different contexts (Lombard & Jones, 2015, p. 14). Depending on the field, the term is understood differently, often leading to confusion and the inability to settle on a unifying theory, which is claimed to threaten our understanding of the concept and its utility. This theoretical ambiguity is problematic in the field of game studies as the concept of presence has been claimed to be particularly relevant to understanding player experience (Calleja, 2011, p. 18).

A central issue in the discourse on presence is its problematic relationship with immersion. Calleja (2011) points out that while the two terms are often used interchangeably, they represent distinct yet interrelated concepts. Immersion has traditionally been framed as an all-encompassing absorption into a virtual environment, whereas presence implies a more nuanced and dynamic engagement, wherein players maintain an awareness of both the game world and their real-world surroundings. This distinction is crucial because it challenges the simplistic notion that immersion requires the complete suspension of disbelief.

Moreover, while immersion is often framed as a binary experience (where one is either immersed or not), presence tends to be conceptualised as a more continuous and multidimensional phenomenon. Rather than relying solely on sensory fidelity or technological realism, presence also encompasses narrative involvement, agency, and social interaction within the virtual world. In this way, presence offers a more flexible framework for understanding engagement, allowing for varying intensities and types of "being there".

For this thesis, I prefer the term presence over others, competing terms, like immersion for different reasons. First, immersion has been used inconsistently to describe divergent aspects of engagement, often conflating many concepts under a single term. This usage makes it difficult to establish a precise theoretical framework. On the other hand, and we will explore this later in the chapter, presence aligns more closely with the spatial and psychological sense of “being there” in a virtual environment. Additionally, presence’s holistic and nuanced approach to player experience makes the concept a better match to understand the factors in play for verticality to influence a player’s connection with a digital game.

Given the significance of presence in shaping player experiences, it is essential to explore the different theoretical perspectives that have been developed to conceptualise and measure it. This chapter provides an in-depth examination of presence theories, discussing their various dimensions and frameworks. By doing so, it lays the groundwork for understanding how verticality in game design contributes to and modifies a player’s sense of presence, a topic that will be further explored in the following chapters.

2.2 Defining Presence

As stated above, the concept of presence has been a recurring theme in numerous fields of study. However, because of its frequent use, its definition remains subject to considerable debate across disciplines. Presence is generally understood as the psychological state in which an individual experiences a sense of “being there” in a mediated environment (Witmer & Singer, 1998). Yet, this seemingly straightforward definition conceals a rich theoretical landscape with multiple interpretations and frameworks attempting to capture the nuances of presence.

2.2.1 Key Definitions and Conceptual Ambiguity

At the beginning of their chapter, Lombard and Jones (2015) identify five distinct definitions of presence to emphasise the diversity in the academic discourse. Waterworth et al. (2015) view presence as a spatial sensation, focusing on the physical experience of being surrounded by an external world. Jacobson (2002), similarly, considers presence in terms of psychological engagement, emphasising the interaction and immersion within a virtual world.

From another perspective, Lombard and Ditton (1997) define presence as the perceptual illusion of nonmediation, where technology becomes “invisible” and creates an experience so seamless with the machine that the user reacts as if it were unmediated reality.

Zahorik and Jenison (1998) introduce yet another dimension, defining presence as being about action and interaction, where the environment responds accurately to one's actions, offering a successful and dynamic experience. Lastly, Cheyne (2001) adds a layer of spirituality and otherness, describing presence as a numinous or supernatural awareness of entities distinct from the self. It is not about ourselves but about the presence of something other than us in the close perimeter.

One of the earliest and most widely cited definitions comes from Witmer and Singer (1998), who describe presence as "the subjective experience of being in one place or environment, even when one is physically situated in another." Their work emphasises the role of sensory fidelity and attentional engagement in creating an immersive experience. Similarly, Biocca (2003) and Calleja (2011) build upon this by conceptualising presence as a dynamic interplay between the user, the medium, and the environment, proposing that presence exists on a continuum rather than as a binary state.

The multidisciplinary nature of the theory resulted in a conceptual ambiguity, often leading to inconsistent usage. For instance, Zahorik and Jenison (1998) interpret presence as "tantamount to successfully supported action in the environment". They approach presence from the perspective of ecological psychology, suggesting that it emerges when a mediated environment provides the necessary perceptual cues for interaction. This ties into Gibson's (2015) notion of perception, where presence is reinforced through the direct perception of actionable possibilities in an environment.

2.2.2 Approach to Presence

Presence is not a monolithic concept but rather one that shifts depending on the discipline or the individual through which it is viewed. In media studies, presence is often linked to the "illusion of non-mediation" (Lombard & Ditton, 1997), where a viewer momentarily forgets the existence of a screen or interface. In psychology, it is examined in terms of cognitive absorption and perceptual realism (Salter & Wilbur, 1997). In game studies, presence is intrinsically tied to, but not limited to, interactivity and agency, as the player's ability to influence the virtual world reinforces their sense of presence (Calleja, 2011; Van Looy, 2003).

The intersection of these perspectives has led to the development of several models attempting to refine presence theory. Wirth et al. (2007) propose the Spatial Presence Model, which suggests that presence is constructed in two stages: the mental representation of a mediated space and the subsequent response to it as if it were real. This aligns with Slater and Wilbur's (1997) distinction between place illusion (the feeling of being in a virtual space) and plausibility illusion (the believability of events occurring within it).

Definitions of scholars like Waterworth et al. (2015), Jacobson (2002) and Lombard and Ditton (1997) provide a general idea of the sensation that presence should feel like, but lacks practicality. Zahorik and Jenison's (1998) dimension of presence provides an interesting addition to the idea of presence as they introduce the need for users to have a feeling of agency in the virtual world. They highlight the fact that a feedback loop (action and interactivity) is essential for enhanced engagement. However, it may not fully account for selfness or narrative.

This lack of consensus in current research poses a challenge for applying presence to specific situations, such as gameplay features like verticality in this thesis. Because the given definitions cannot provide a precise and clear perspective on the notion but only vague and mostly abstract ideas, the concept of presence is inaccessible for applications unless it is worked on and clarified for the specific use. For this reason, it is crucial that the notion of presence and its definitions must be operationalised clearly for a practical application. This issue was identified by Calleja (2011), who tries to reframe the concept of presence through his player involvement model. This framework positions presence as a usable concept for games compared to purely perceptual definitions.

2.3 Dimensions and Categories of Presence

Because presence is a multi-dimensional concept, some scholars suggest a framework to categorise the concept into different dimensions. Lombard and Jones (2015) offer a comprehensive classification that includes spatial, social, and self-presence, as well as emotional engagement, perceived realism, and cultural presence. These dimensions help illuminate how presence manifests in different contexts and how it contributes to the experience.

One of the most widely discussed aspects is *spatial presence*, which refers to the perceptual illusion of physically "being there" in a mediated environment despite knowing that the person is actually elsewhere. This sensation emerges when an individual perceives a digital space as an extension of their real-world surroundings. Researchers like Witmer and Singer (1998) describe spatial presence as a psychological state where a person's focus shifts from their actual environment to the virtual one. Another term for this is telepresence since it occurs when individuals feel physically present in a remote or virtual space (Biocca et al., 2003). It is often discussed in the context of virtual reality or telecommunication technologies.

Occasionally, scholars distinguish between the sense of "being there" in a virtual environment (Saari, 2004) and "being there" in a real and remote location of the actual world (Minsky, 1980).

Another dimension is *social presence*, which refers to the feeling of being with others within a mediated environment. Whether they are real people (such as other players in an online game) or artificial agents (such as NPCs or AI-driven characters), users often experience a sense of mutual awareness and interpersonal connection. Lombard and Ditton (1997) emphasise that social presence extends beyond mere co-location. It involves a sense of mutual awareness and interaction, where users feel that virtual characters or other individuals are "real" and responsive.

Beyond the presence of others, *self-presence* captures how individuals experience a sense of their own identity, emotion and body when engaging with the technology. It encompasses how the users perceive themselves in the virtual environment, particularly in relation to avatars or other representations of themselves. The degree to which users identify with their virtual representation influences their sensation of presence.

Lombard and Jones (2015) highlight another facet of presence they call *engagement*. This category focuses on psychological engagement, synonymous to involvement and closely linked to concepts like attention. It refers to the depth of cognitive and emotional connection an individual experiences within a mediated environment. Similar to Lombard and Ditton's (1997) Presence as Immersion, a user is "involved, absorbed, engrossed by stimuli from the virtual environment". The stronger the engagement, the less aware an individual becomes of their real-world surroundings, further reinforcing the sensation of presence.

This idea of a multi-faceted notion of presence is further illustrated by the coexisting notion of *transportation*, which refers to the psychological sensation of being "carried away" into another world, a phenomenon associated with deep narrative engagement. Green and Brock (2000) describe transportation as the process of becoming deeply absorbed or immersed in a narrative world, leading them to feel as if they are part of the unfolding events rather than merely observing them, the feeling of being lost in a story. It involves multiple interrelated components. First, it includes imagery and mental visualisation, where the "transported" form vivid mental pictures of the story's events and characters. This is accompanied by emotional engagement as the narrative elicits strong emotional responses, enhancing the believability of the events. Additionally, transportation requires a high level of attentional focus, during which individuals become so engrossed in the story that they may lose awareness of their surroundings and real-world distractions. Another key aspect is psychological distancing, where individuals temporarily set aside their own beliefs or perceptions, becoming more open to the influence of the narrative. Together, these elements make transportation a convergent mental process that enables readers to connect deeply with the story's content and characters, which can also lead to changes in beliefs and evaluations related to the narrative.

In game studies, Calleja (2011) discusses player involvement as a dynamic process where different layers of engagement, ranging from mechanical interactions to narration, contribute to the overall experience of presence. Thus, he argues that engagement should not be a distinct dimension of presence, but rather a prerequisite for a player's whole presence sensation when playing a digital game.

Another dimension that is interpreted is *perceived realism*. This category describes how convincing and lifelike a mediated experience appears. Lombard and Jones (2015) highlight that realism is not solely about high-fidelity graphics or sensory accuracy. It is rather about coherence and plausibility within the virtual environment. Even stylised or abstract virtual worlds can evoke a strong sense of presence if they maintain consistency and afford believable interactions.

An additional dimension of presence is cultural presence, introduced by Lombard and Jones (2015). A dimension where Mantovani and Riva (1999) challenge the binary between "real" and "virtual" by suggesting that presence is shaped by biocultural processes. They argue against the notion that "real" objects exist independently of human minds and "virtual" objects exist solely within them. Instead, they propose that human experiences are bioculturally mediated. This means there is no strict separation between external objects and internal perceptions. From this standpoint, presence is defined as the ability of individuals to recognise themselves, others, and objects not only as existing within a physical space but also as part of a complex sociocultural network that connects people, objects, and their interactions.

Lombard and Jones (2015) note that Mantovani and Riva's (1999) idea appears to underpin all other dimensions of presence, as it emphasises the inevitable influence of language and culture on any experience of presence. They also identify one last dimension that they call "parapresence", which focuses on the perception that a person is physically present in the environment even though they are not and could not logically be there (e.g. a widow sensing her husband's presence). However, this dimension moves beyond the scope of digital games and will not be pursued further in this work.

By distinguishing these dimensions, Lombard and Jones (2015) argue that researchers can better analyse the various ways presence manifests across different media. Whether it emerges through physical navigation in virtual space, social interaction, emotional absorption, or narrative engagement, presence is a complex phenomenon, and these categorisations provide a foundation. For this thesis, highlighting these categories illustrates that various factors influence presence and cannot be defined by a sole dimension. Fortunately, in game studies, Calleja (2011) highlighted this situation and conceptualised a framework for a comprehensive approach toward presence.

2.4 Presence in Game Studies

The study of presence within digital games is unique when compared to other media, such as books or televisions. Indeed, digital games require interaction, agency, and decision-making. Players are active participants in the construction of their experience. Among the scholars exploring presence in games, Gordon Calleja's (2011) model of player involvement offers a nuanced understanding of player experience. Additionally, the role of space in presence further highlights the importance of environmental affordances and player movement in deepening player experience. Calleja suggests that players experience games through a gradual process of involvement, where presence emerges as a result of habitual and embodied interactions with the game world rather than as an abrupt shift into a fully immersive state. By reframing presence as a process of engagement rather than an absolute state, Calleja offers a more flexible and context-sensitive approach to understanding players' connection with a virtual environment. This perspective acknowledges that presence is fluid and multifaceted.

2.4.1 Player Involvement

To address the challenges revolving around presence research, Calleja (2011) addresses involvement and establishes a model that illustrates different categories of involvement. Because involvement is an ingredient that fosters the experience of higher cognitive states like presence, it coheres with the idea of exploring its multiple facets. The author breaks down player involvement into six dimensions, each contributing to the broader experience of presence.

1. Kinesthetic Involvement

In Calleja's player involvement model (2011, pp. 55-71), kinesthetic involvement refers to the player's physical interaction with the game through controls and movement, forming a key sensory-motor relationship that grounds the experience of presence. Kinesthetic involvement is rooted in the immediate, physical interaction between the player and the virtual environment. What Calleja terms the "micro-level" of moment-to-moment interaction.

This type of involvement relies on how fluid and responsive the control mechanisms are. The tactile feedback from a gamepad, the ease of translating intention into action through button presses, or the precise input of mouse movements all contribute to how connected the player feels to their avatar or in-game entity. When the game responds seamlessly to the player's input, it fosters a sense of embodied agency. The player doesn't just control the character, they feel part of it.

Calleja draws attention to the importance of learning curves and motor familiarity for players to feel engaged. As players become more familiar with the control scheme, their actions shift from conscious decision-making to automatic, bodily reflexes. This creates a feedback loop where the interface fades, and interaction feels natural and intuitive. This is similar to the concept of “transparent immediacy” discussed by Bolter and Grusin (1999), where the player no longer notices the mediation and instead feels directly present in the virtual world.

Importantly, kinesthetic involvement isn’t just about raw responsiveness. It’s also about the feel of the movement: weight, inertia, resistance, momentum. These elements contribute to how players perceive the physical properties of objects within a game and how those properties affect gameplay (Swink, 2008, p. 169). For example, Steve Swink (2008) analyses *Gears of War* (Xbox Game Studios, 2006) based on these elements of physics to define how the game shapes a player’s perception of the physical reality of the game world.

From Calleja’s perspective, kinesthetic involvement plays a foundational role in fostering presence. While narrative and affective elements deepen a player’s investment, it is often through kinesthetic engagement that players first ‘enter’ the game world. Without a satisfying control experience, the experience of a player can quickly break down. Thus, kinesthetic involvement acts as a gateway. Moreover, kinesthetic involvement varies not only by genre or gameplay mechanics but also by hardware platform. Motion controllers, haptic suits, or VR systems impact this type of involvement by trying to create more immediate physical correspondence between real-world motion and virtual action.

In 3D open-world digital games, such as *Watch Dogs: Legion* (Ubisoft, 2020) or *The Legend of Zelda: Tears of The Kingdom* (Nintendo, 2023), there is an additional layer of complexity and, arguably, richness to kinesthetic involvement. Verticality expands the axes of movement beyond the horizontal plane. While moving left, right, forward, or backwards forms the basis of spatial navigation, the ability to move upward and downward, whether by climbing, jumping, gliding, or flying, requires a more varied set of motor inputs and sensory coordination. For example, in *Watch Dogs: Legion* (Ubisoft, 2020), players are able to use their character to call a drone on which they can get on. This allows the player to explore and get access to locations that would not be accessible simply on foot.



Controlling a drone in *Watch Dogs: Legion* (Ubisoft, 2020)

Calleja's concept of kinesthetic involvement emphasises the micro-level engagement of the player with the game world through physical input. Vertical movement mechanics heighten this engagement by introducing new challenges and sensations that disrupt routine locomotion. For example, scaling a mountain in *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017) demands the player to plan and pay close attention to the stamina wheel. Another example would be coordinating jumps with your partner in playing a parkour section in *It Takes Two* (Electronic Arts, 2021) in order to progress through the story. These actions, as Calleja (2011, p. 137) notes, intensify the player's bodily engagement by requiring continuous micro-adjustments that give a sense of embodied interaction.

Moreover, verticality requires players to develop new motor fluency. Learning how to control gliding movements or latch onto climbing surfaces like in *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017) or even use Rico's grappling hook in *Just Cause 3* (Square Enix, 2015) all contribute to the kinesthetic learning curve. As Calleja notes, the more these controls become second nature and less demanding, the more the interface disappears, allowing for a deeper engagement.



Climbing in *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017)

In this way, verticality becomes an important amplifier of kinesthetic involvement. It enriches the player's bodily interaction with the virtual world, makes movement more varied and expressive, and strengthens the sensation of being inside a dynamic, explorable space. In games where verticality is central, such as *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017), *Horizon Zero Dawn* (Sony Interactive Entertainment, 2017), or *Titanfall 2* (Electronic Arts, 2016), kinesthetic engagement is inseparable from vertical design. The sensation of ascending, falling, soaring, or perching high above the landscape reinforces presence as a bodily and spatial phenomenon.

2. Spatial Involvement

The second dimension of involvement introduced by Calleja (2011, p. 43) is called spatial involvement. In his model, Calleja explains that spatial involvement refers to the player's engagement with the layout, structure, and navigability of the game world. It is the category of involvement that describes how players come to perceive and mentally map the spaces they move through. Unlike kinesthetic involvement, which is concerned with the physical interface between the player and the avatar, spatial involvement develops over time as players become familiar with the geography of the virtual environment, learning to navigate its terrain, landmarks, and boundaries.

Calleja (2011, p. 92) notes that spatial involvement emerges through the player's progressive internalisation of the game world. This process is not immediate but is built through repeated

exposure, exploration, and interaction with the virtual space. At first, players may rely heavily on maps, markers, and HUD elements to navigate. However, as they begin to form mental maps and recognise environmental cues, they start navigating more intuitively. This idea of cognitive mapping is a concept introduced by Edward Tolman (1948).

A key part of spatial involvement is how games communicate their navigational logic. Level design, environmental affordances, and spatial cues all influence how easily players can make sense of and orient themselves within the world. Games use visual landmarks, spatial rhythms, and distinct topographies to help players create mental maps. For instance, *Dark Souls* (FromSoftware, 2011) uses towering structures and interlocking shortcuts which takes part in building an understanding of the world's physicality and scale. Players internalise the digital space with practice.

Calleja, in his tenth chapter (2011, p. 167), makes a crucial distinction between simply viewing a space and incorporating it into one's lived experience. A rich spatial involvement is the result of this incorporation. Players feel a deep connection with the digital world not only because the space looks realistic or is detailed, but because they come to understand and navigate it fluently. This fluency reinforces the sense of presence. Incorporation as termed by Calleja, is defined as the moment when players inhabit the game world through habitual, embodied interaction.

Moreover, spatial involvement is also affective and experiential. The feeling of standing on a cliff's edge, overlooking a vast landscape in *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017), is both spatial and emotional. The terrain itself shapes the player's psychological experience and engagement. In such moments, verticality becomes a powerful extension of spatial involvement. The ability to traverse high and low spaces adds dimensional depth to the environment, making it feel more tangible and explorable. Importantly, spatial involvement evolves over time. As players move from novice to expert, their perception of space shifts. What was confusing or maze-like becomes familiar and more coherent. This temporal development is central to Calleja's model, which views presence as something built through practice rather than instantly achieved through high-fidelity graphics or sensory input.

Lastly, the author adds that spatial involvement can be influenced by social contexts in multiplayer games. Reactions from other players, like calls for help or collaborative strategies, blend in with the player's navigation experience. Especially because in multiplayer settings, good spatial knowledge tends to provide a competitive advantage. These factors have an impact on a player's engagement with their digital space.

3. Shared Involvement

Another key component of Calleja's (2011) Player Involvement Model is shared involvement, which refers to the engagement that arises when players are aware of and interact with others, whether real players or computer-controlled characters, within the game environment. This type of involvement closely aligns with the concept of social presence, as defined by Lombard and Ditton (1997), where the presence of others in a mediated environment contributes to the user's sense of "being there". In games that support multiplayer or socially dynamic systems, such as massively multiplayer online games (MMOGs) like World of Warcraft (Blizzard Entertainment, 2005) or Elder Scrolls Online (Bethesda Softworks, 2014), simultaneous cohabitation of virtual space with others reinforces a player's shared involvement, creating a persistent social ecosystem.

Shared involvement can take many forms, including cohabitation (simply being in the same space as others), cooperation (working together toward shared objectives), or competition (engaging in rivalry and challenges). The social context of these interactions shapes the player's experience. For instance, knowing that other players are present in the same environment may alter how one navigates the world, approaches missions, or responds to events. For example, in Minecraft (Mojang Studios, 2011), I would not have the same strategy for establishing a base and gathering resources depending on whether I am in a Player versus Player (PvP) server or simply playing alone. I might decide to be careful and avoid players by changing my route, or, on the contrary, I could see other players as an opportunity to improve my weapons and objects. The presence of others transforms the game from a solitary task to a socially embedded experience. These interpersonal dynamics impact decision-making. According to Lombard and Jones (2015) on social presence, these social layers contribute to a more complex and believable perception of the game world. Hence, fostering shared involvement in a player.

The development of social ties is another essential dimension of shared involvement that Calleja introduces. MMOGs often foster friendships and informal communities that extend beyond the game itself. Players build these relationships through sustained interaction and shared experiences, such as completing quests in Thrones and Liberty (Amazon Games, 2024) or simply inhabiting the same virtual spaces as in Minecraft (Mojang Studios, 2011). These recurring engagements lead to familiarity and trust that resemble real-world social conventions. The informal norms of everyday life, such as greetings, etiquette and cooperation, are easily transplanted into the gaming context, enhancing the player's connection with the virtual world (Taylor, 2006).

The role of persistence in game environments is central to the formation of shared involvement. Unlike traditional multiplayer sessions that dissolve when a match ends, persistent worlds allow for ongoing interaction. Players see each other's avatars inhabiting the world even when not engaged in

direct collaboration. This continuity helps foster a perception of a living, breathing world. A world in which the player is not alone, but rather part of an active, evolving community. This persistent cohabitation heightens engagement and reinforces the player's perception of a socially situated involvement (Calleja, 2011; Schroeder, 2002).

The social nature of the environment also directly influences gameplay. Players adapt strategies in real time, respond to social cues, and often form temporary or long-term alliances. Quests may be approached differently when undertaken in groups, and competitive scenarios are shaped by rivalries and team dynamics. These social interactions are additive to the experience, and integral to how the game is played and understood. The influence of others heightens emotional engagement, as players share triumphs, defeats, and unexpected outcomes together, creating affective ties that further deepen the presence experience (Isbister, 2016).

From a sociological perspective, the types of relationships formed in these environments can be understood through the concept of social capital. MMOGs tend to foster “bridging” social capital (broad, casual connections with many players) rather than “bonding” capital, which is characterised by deep, emotionally supportive relationships (Putnam, 2000). While the fast-paced and expansive nature of online games may not immediately lend itself to deep intimacy, the accumulation of shared experiences can nonetheless lead to lasting connections. These social ties, even if weak at first, contribute to the emotional texture and memory of the game world, helping players feel anchored within a social landscape.

Notably, shared involvement is not exclusive to multiplayer games. Single-player titles can simulate meaningful social engagement through the use of well-designed AI companions. Characters like Ellie in *The Last of Us* or Chumbucket in *Mad Max* serve not only mechanical purposes but also narrative and emotional ones. Their presence contributes to the player's sense of not being alone, as they react, converse, and participate in the game's world in responsive and believable ways. These companions support a form of simulated social presence, which, while not emergent from real social interaction, still reinforces the sensation of shared experience.

4. Narrative Involvement

The fourth dimension of involvement given by Calleja is narrative involvement (2001, pp. 113-133). The approach regarding narrative involvement is to look at how stories are experienced in digital games. Because the discussion of narrative in game studies has been significant (Aarseth, 2012; Eskelinen, 2004; Juul, 2001), it is essential to acknowledge the impact of stories in a player's gaming experience.

Narrative involvement, within Calleja's (2011) Player Involvement Model, refers to the player's engagement with the events, characters, and unfolding storylines within the game world. This form of involvement captures both emotional and cognitive investment in narrative structures and reflects the player's sense of participating in a meaningful, evolving storyline. These narrative structures could be linear, branching or emergent. Unlike media such as literature or film, where the audience passively receives a predetermined sequence, digital games allow players to influence, interpret, and often co-create the story through their actions. This interactivity transforms narrative into a lived experience, that has a non-negligible influence on the experience (Calleja, 2011, p.170).

Calleja emphasises that narrative involvement is not limited to scripted plotlines or cutscenes (2011, p.119). Instead, it encompasses a wide array of storytelling techniques, including environmental storytelling, dialogue systems, and player choice (p. 124). The story in many games is embedded within the space itself, through visual cues, environmental details, and/or the structure of the world. Games like Bioshock (2K, 2007) and The Last of Us (Naughty Dog, 2013) exemplify this integration, offering players not only a plot to follow but a world that tells its own story. In such cases, exploration becomes an act of narrative discovery, impacting emotional resonance and spatial engagement simultaneously.

Crucially, narrative involvement in games operates both temporally and spatially. Temporally, players are drawn into a sequence of events that unfolds over time, shaped by their decisions and actions. Spatially, the world itself often functions as a narrative artifact: architecture, objects, and level layout can all provide story context, historical depth, or foreshadowing. As Henry Jenkins (2004) has argued, games construct narrative architecture, where story and space are inseparably linked. This design philosophy allows players to engage with the narrative not just as observers, but as active participants who interpret and construct meaning as they move through the environment.

Narrative involvement also plays a key role in emotional or affective presence. When players develop attachments to characters, struggle with moral dilemmas, or experience moments of triumph and loss, they become emotionally embedded in the game world. These responses are not merely reactions to

external stimuli, but internalised experiences that contribute to a deeper sense of being “in” the story. As Calleja notes, such involvement is not necessarily linear or dependent on the main plot — players may find emotionally meaningful moments in side quests, personal choices, or even emergent gameplay scenarios that create unplanned but memorable narratives.

Furthermore, narrative involvement exists on a spectrum of how the story is conveyed in the game, depending on both the design of the game and the player’s interpretive engagement. Some games offer tightly controlled narrative experiences, such as *The Last of Us* (Naughty Dog, 2013), where the story is central and progression is guided. Others, like *The Elder Scrolls V: Skyrim* (Bethesda, 2011), provide expansive worlds in which players encounter story elements non-linearly, shaping a personal narrative through exploration and interaction. Even in sandbox games like *Minecraft* (Mojang Studios, 2011), players construct their own narrative experiences through self-directed goals and creative play. These emergent narratives reflect what Janet Murray (2017) calls “procedural authorship”, where players generate meaning through interaction with systems and rules rather than through fixed scripts.

Importantly, Calleja does not treat narrative involvement as isolated from the other dimensions. It is deeply entangled with other categories of involvement. For instance, the physical act of traversing a treacherous cliff may be intensified by narrative context. Perhaps the climb represents a critical turning point in the story or a moment of character development. In *Shadow of the Colossus* (Sony Computer Entertainment, 2005), for example, the act of scaling each towering creature carries not only mechanical challenge but emotional weight, as players gradually realise the moral cost of their quest. In such cases, narrative amplifies its emotional and symbolic significance.

However, it is important to consider narrative involvement as a fragile variable. It may be diminished if the story lacks coherence, if players are alienated by unrelatable characters, or if player agency is undermined by overly scripted experiences or too many cutscenes. Also, players differ in how much narrative they seek but also how much they attend to (e.g. I will not experience as much narrative during my gameplay session if I do not focus on the story elements), and, as we have seen, there are many possibilities for storytelling, from a traditional cutscene to telling a story with the environment. This variability underscores the importance of designing for different narrative appetites and acknowledging that narrative involvement is one of many possible starting points into presence.

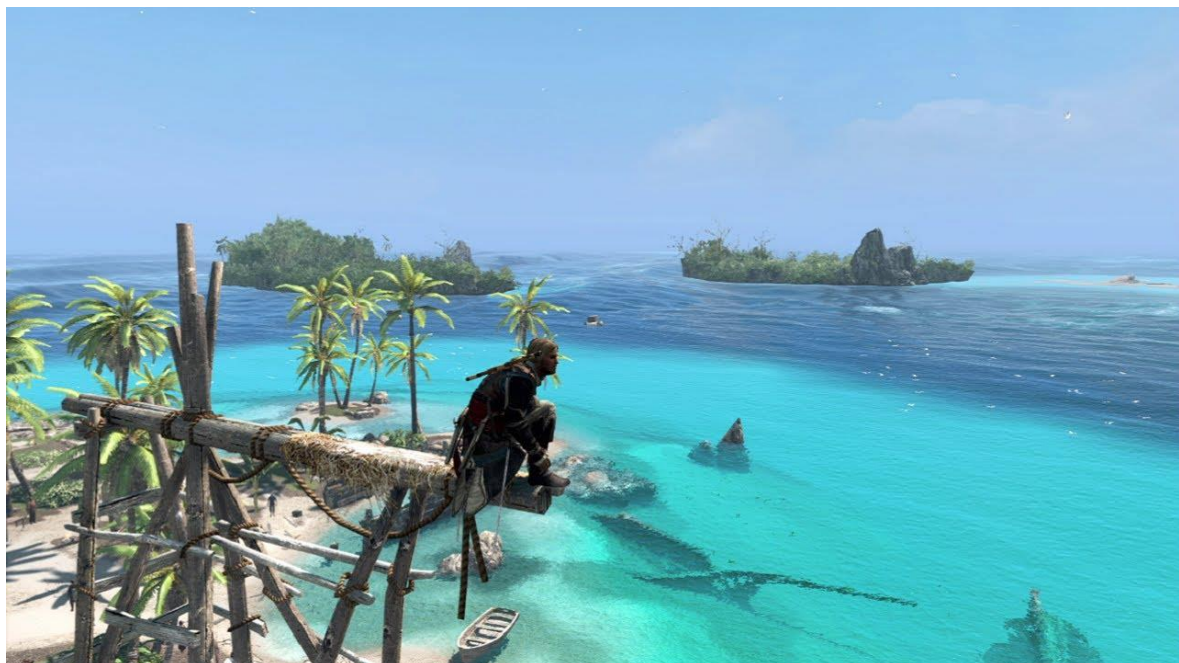
5. Affective Involvement

When players develop a psychological investment in the game’s world, characters, or events, an emotional connection materialises. Horror games like *Resident Evil* (Capcom, 1996-2023) or *Silent Hill*

(Team Silent, 1999) use tension, atmosphere, and uncertainty to heighten affective involvement in an attempt to make the experience feel psychologically immediate. Similarly, emotionally impactful storytelling, such as the final moments of *Shadow of the Colossus* (Sony Computer Entertainment, 2005), can provoke strong emotional responses from players.

Affective involvement refers to the emotional engagement players develop in response to in-game events, characters, and environments. Calleja (2011) identifies this dimension as a central component of the player experience, rooted in both moment-to-moment emotional reactions (micro-involvement) and longer-term affective investment (macro-involvement) with the game world.

At its core, affective involvement is shaped by a cybernetic feedback loop between the player's input and the game's response, creating a dynamic interaction that amplifies emotional engagement. Calleja argues that this interplay makes digital games a particularly potent medium for generating emotional intensity (often more so than non-interactive media) because players are both authors and audience in their experience (2011, pp. 135–136). This loop allows players to experience a broader emotional spectrum, ranging from exhilaration in combat, fear in horror titles, to contemplative admiration when encountering breathtaking landscapes (e.g. the beautiful Caribbean archipelago landscape view when climbing to a high location in *Assassin's Creed IV: Black Flag* (Ubisoft, 2013))



Sync location in *Assassin's Creed IV: Black Flag* (Ubisoft, 2013)

The aesthetic and narrative design of game environments often plays a rhetorical role in eliciting specific emotional responses. However, Calleja notes that these intended emotions are not always what players experience. Emotional resonance depends on a player's disposition, attention, genre preference, or even their interpretation of game events (2011, p. 140). For example, a deep dive in the Mediterranean sea in *Assassin's Creed Odyssey* (Ubisoft, 2018) may evoke peace in one player thanks to its details and unease in another because of the possible presence of danger. It depends on personal associations and gameplay context. This highlights affective involvement as a deeply subjective and interpretative experience.

Moreover, Calleja draws attention to excitatory homeostasis (2011, p.135), a concept from media psychology suggesting that individuals seek optimal levels of arousal through entertainment. Games, through their interactive nature, provide the tools for players to calibrate emotional stimulation. For instance, overstimulated players may turn to serene games like *Journey* (Thatgamecompany, 2012) or *Stardew Valley* (ConcernedApe, 2016) for calming experiences, while others may seek adrenaline in fast-paced shooters or horror games (p. 137).

Designers often leverage these emotional hooks to enhance affective involvement. A compelling example is the rhythmic tension and release in survival horror games like *F.E.A.R. 2: Project Origin* (Warner Bros. Interactive Entertainment, 2009) or *Resident Evil* (Capcom, 1996-2023), which juxtapose intense combat with eerie calm to amplify emotional reactions (Calleja, 2011, p. 144). On the other hand, the emotional impact can also arise from narrative moments, such as loss, betrayal, or moral decisions, that contribute to the player's emotional investment in the game's world and characters. A famous example is the betrayal scene in *Call of Duty: Modern Warfare II* (Activision, 2022), where Shepherd, a key ally character in the story, deceives the player by eliminating them.

Importantly, just like the other categories, affective involvement does not exist in isolation. Calleja emphasises the interplay between affective involvement and other dimensions, particularly narrative and spatial involvement. The emotional resonance of a game scene is heightened when it is embedded in a meaningful narrative context or set within a space that the player has come to perceive as familiar or symbolic (Calleja, 2011, p. 146). Thus, affect is not only a product of game stimuli but also of the player's evolving relationship with the world and its agents.

6. Ludic Involvement

The final dimension of involvement given by Calleja (2011) is called ludic involvement. Ludic involvement refers to the player's engagement with a game's rule systems, goals, and decision-making

processes. It is the cognitive and strategic dimension of gameplay, where players interact with challenges by forming plans, evaluating choices, and adapting to feedback. While kinesthetic involvement focuses on the physical execution of actions, ludic involvement centres on the mental engagement required to navigate systems of rules and objectives (p. 147). This interaction fosters a sense of agency, encouraging players to actively participate in shaping their gameplay experience. This is a key contributor to the overall sense of presence in virtual environments.

Calleja divides ludic involvement into two phases: the macro phase, where players pursue long-term goals or aspirations (such as completing a campaign or reaching a certain level), and the micro phase, which includes moment-to-moment problem-solving and in-the-moment decision-making (2011, p. 150). In the macro phase, engagement is guided by overarching motivations (progression, mastery, achievement) that help structure and sustain player activity over time. A case in point is *World of Warcraft* (Blizzard Entertainment, 2005), where players may set personal milestones like acquiring elite equipment. Objectives that shape daily interactions and long-term play (p. 155).

The micro phase, by contrast, occurs on the immediate, tactical level. Players evaluate options, make fast decisions, and execute strategies in response to the evolving demands of the game. According to Calleja, these layers of activity operate within a goal hierarchy, where systemic goals defined by the game's rules coexist with self-imposed objectives defined by the player (2011, p. 152). For example, a player in *Grand Theft Auto IV* (Rockstar Games, 2008) might alternate between pursuing narrative missions and deciding to explore the world freely in search of a new vehicle. Both activities fall within the scope of ludic involvement, as they require active engagement with the rules and logic of the game world.

To articulate this decision-making structure, Calleja introduces a framework of strategies, tactics, and microtactics. Strategies encompass high-level planning over extended sessions, tactics are short-term plans tied to particular scenarios, and microtactics describe reactive decisions made in real time (2011, p. 157). In a tactical shooter like *The Finals* (Embark Studios, 2023), for instance, a team may develop a strategy for zone defense, coordinate specific tactics for room entry, and rely on microtactics for moment-to-moment gunfights and movement decisions. This layered system of planning and adaptation is central to ludic engagement and shapes how players experience success and challenge.

Importantly, ludic involvement is not exclusive to competitive or combat-driven games. It also includes more open-ended or creative forms of interaction. In sandbox games like *The Sims 4* (Electronic Arts, 2014) or *Minecraft* (Mojang Studios, 2011), players often establish self-directed objectives, such as building a structure or optimising a farm. These player-authored goals involve engaging with the

affordances and constraints of a structured rule system, even if the challenge is not externally imposed.

Presence, in this context, is reinforced through successful mastery of the game's logic. As players make decisions and receive clear feedback (e.g. rewards), they become more embedded in the game's internal systems, experiencing what Csikszentmihalyi (1990) would describe as a flow state, a balance of challenge and skill that leads to deep concentration and loss of self-consciousness. This state of optimal engagement is deeply aligned with ludic involvement, where understanding and controlling the game's logic strengthens the player's belief in the coherence and reliability of the virtual world. As Calleja notes, this feedback loop (of intention, action, and system response) reinforces the feeling of acting within a believable and reactive environment (2011, p. 151).

It is also worth noting that different genres stimulate ludic involvement to varying degrees. Strategy, puzzle, and roguelike games typically demand more sustained attention to mechanics and planning, whereas narrative-driven or exploration-based games may distribute cognitive load across multiple dimensions of involvement. Still, even in the latter, ludic structure remains fundamental: whether solving a riddle in *Return of the Obra Dinn* (3909 LLC, 2018) or a dungeon in *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017), the logic of gameplay must be understood and mastered.

Dynamic Interplay Between These Dimensions

Calleja (2011) emphasises that the six dimensions of player involvement do not operate in isolation. Rather, presence emerges from their dynamic interplay, where engagement in one dimension often supports or intensifies engagement in another. The experiential depth of gameplay emanates from this fluid and context-sensitive blending. A player exploring an open-world environment may experience kinesthetic and spatial involvement through movement, while simultaneously engaging in narrative and affective involvement if the environment is tied to story elements or emotional stakes. Games that successfully align multiple forms of involvement (e.g. *Half-Life: Alyx* (Valve, 2020), which integrates VR controls, environmental storytelling, and spatially responsive design) foster a continuous flow for the player to feel engaged.

Kinesthetic and spatial involvement often feel closely connected. As players move through a game world, the tactile and responsive qualities of movement (its weight, precision, and fluidity) fundamentally shape how space is perceived and navigated. A disjointed movement system can render even the most detailed environments disengaging, whereas fluid mechanics allow players to internalise the world's structure and rhythm. Bodily interaction thus informs spatial cognition,

anchoring the player within a world that feels consistent and responsive. This interplay is especially relevant in vertical spaces, where climbing, gliding, or falling plays in altering spatial orientation.

Narrative and affective involvement likewise can reinforce one another. Narrative events often act as emotional triggers, evoking responses such as empathy, awe, or moral tension. In turn, emotionally charged experiences increase the narrative importance of gameplay moments. Ryan (2015) argues that interactive environments offer narrative meaning through experiential engagement, enabling players to live the story rather than consume it passively. Games such as *The Last of Us* (Naughty Dog, 2013) or *Life is Strange* (Square Enix, 2015) exemplify this relationship, where decisions and characters are tightly linked.

Shared involvement often acts as a mediator between affective and ludic engagement. The presence of others transforms objectives into social encounters. Cooperative missions, player versus player (PvP) zones, and exploration introduce interpersonal dynamics (strategy, emotional investment, etc.). In massively multiplayer online role-playing games (MMORPGs) such as *The Elder Scrolls Online* (Bethesda Softworks, 2014), for instance, players often tailor their gameplay strategies based on group composition and social relationships. In *Fallout 4* (Bethesda Softworks, 2015), a single-player role-playing game, the player will have to strategise their attack based on the type of enemy and their companion they chose, since each have particular skills.

We can also look into the synergy between ludic and narrative involvement. Game rules and systems are channels for challenge and reward but also act as thrusters for progression in the narrative. In *Disco Elysium* (ZA/UM, 2019), dialogue choices and moral dilemmas are embedded in the game's design. Players shape the unfolding story through thinking, becoming co-authors of the narrative by acting.

Calleja (2011, p. 161) notes that the importance of each involvement dimension fluctuates dynamically throughout gameplay, depending on the player's intentions, the game's design, and the immediate context. A single session may involve transitions between kinesthetic action, narrative sequences and emotional bounds with the in-game avatars. These shifts overlap and feed into one another, creating an ongoing feedback loop between the player and the machine that sustains presence. Crucially, this interplay is the foundation of incorporation (Calleja, 2011, p. 169).

Conversely, when these dimensions are out of sync, presence can deteriorate. Mechanically repetitive tasks may undermine narrative momentum, inconsistent environmental design may obstruct spatial orientation, and shallow social mechanics may hinder emotional connection. As Calleja warns,

presence is fragile, it must be actively maintained through balance and coherence across all levels of involvement (2011, p.172).

2.4.2 The Relationship Between Spatial Navigation and Presence

While Calleja's (2011) model outlines a comprehensive framework for player involvement, one particular aspect that strongly connects to verticality is how spatial navigation influences a player's sense of presence, which is demonstrated in chapter 4. Movement through space is a central vector of player engagement which relates to spatial and kinesthetic involvement. Moreover, spatial involvement explains how the way a game enables players to traverse its environments can significantly shape how that environment is understood, felt, and inhabited.

As I discussed in section 2.4.1, spatial involvement is stimulated when players form a cognitive map of the game world through active engagement, gradually internalising its architecture and geography. This process is reinforced through kinesthetic feedback: walking, climbing, jumping, gliding, or flying are tactile expressions of how players relate to space. The responsiveness and physicality of traversal deeply influence the experience. For instance, mechanics like parkour (*Dying Light*, Techland, 2015), jetpacks (*Titanfall 2*, Electronic Arts, 2016), or grappling hooks (*Just Cause 3*, Square Enix, 2015) create a sense of physical engagement with the game world. Here, kinesthetic and spatial involvement converge to support presence.

Consequently, to understand the factors influencing the role of verticality in player experience, it is worth theorising what space itself is. How it is constructed, imbued with meaning, or perceived culturally and psychologically. This leaves open foundational questions: What mental and perceptual processes underpin spatial involvement? How do virtual environments simulate, reinforce, or deviate from real-world notions of space and place? And crucially for this thesis: how does vertical movement through layered, three-dimensional environments reshape the player's spatial understanding?

These questions are vital for understanding the role of verticality in games. Vertical navigation such as ascending a tower, gliding from a cliff, descending into a dungeon, reshapes spatial orientation, challenges cognitive maps, and arguably creates affordances for player agency (see section 4.2). Gibson (1979) describes the concept of affordance as the way an environment provides actionable opportunities based on the player's capabilities. Open-world games that grant freeform exploration, such as *Just Cause 3* (Square Enix, 2015) or *The Legend of Zelda: Tears of the Kingdom* (Nintendo, 2023), allow players to develop a deeper connection with their surroundings by actively engaging with the terrain. To explore this further, it is necessary to move beyond experiential models and into

conceptual frameworks that interrogate the nature of space itself. The next chapter will therefore examine spatial theory. These perspectives will offer a theoretical backbone for analysing how verticality is designed, perceived, and understood in digital environments, and how it contributes to a player's sense of presence.

Chapter 3: Theories of Space, Place, and Perception

The previous chapter explored the concept of player presence and how it is shaped by a dynamic interplay between the six dimensions of involvement. At the heart of this interplay lies space, as a playground for users, but also as a construct that is navigated, and interpreted by individuals. In digital games, it is a constructed environment through which players move, act, and experience. For example, wandering through the plains of *Shadow of the Colossus* (Sony Computer Entertainment, 2005), climbing towers in *Assassin's Creed II* (Ubisoft, 2009), or driving through the dense urban streets of *Cyberpunk 2077* (CD Projekt, 2020). Players engage with space and spatiality is used to shape gameplay, evoke emotion, convey narrative, and deepen player agency (Jenkins, 2004; Calleja, 2011).

To better understand how spatial experience contributes to presence and engagement, and how verticality emerges as a critical design feature, I now turn to broader theoretical perspectives. This chapter lays the groundwork for a conceptual understanding of space, place, and perception, drawing from various disciplines to reframe how space functions within digital games. It begins with the work of Yi-Fu Tuan (1998; 2001), who distinguished between space as an abstract expanse and place as space made meaningful through human experience and attachment. Alongside Tuan, scholars such as Edward Relph (Seamon & Sowers, 2008) and David Canter (1977) have explored how individuals develop psychological, emotional, and behavioral bonds with environments, the "sense of place". These theories offer a phenomenological perspective on how spaces acquire value and familiarity.

Spatial understanding also depends on internal cognitive processes. Kevin Lynch's *The Image of the City* (1960) introduces the concept of imageability, arguing that cities, and by extension, digital environments, become more engaging and navigable when they include legible elements like paths, edges, nodes, and landmarks. This work has been foundational in urban design, architecture, and game studies (Brouchoud, 2013; Yeleswarapu, 2021). This highlights how space fosters engagement. Furthermore, cognitive mapping theories (Tolman, 1948) and neuroscientific research on spatial memory (Maguire et al., 1998) support the idea that individuals form mental representations of space over time, constructing a relationship with memory.

Yet space is not only shaped by cognition and perception, it is also socially and culturally produced. Henri Lefebvre (1991) emphasises that space is embedded with ideological, symbolic, and functional meaning. His model offers a lens through which to interpret spatial storytelling and environmental semiotics in games. Game worlds, like cities, are designed with intention, embedded with values, and experienced through social norms and affordances (Nitsche, 2008). Designs, such as gates, barriers, or vertical pathways, guide and build limitations, tell a story, and can suggest power structures within the world (e.g. Diamond City in *Fallout 4* (Bethesda, 2015) which segregates the different social classes by placing the rich individuals and the mayor's office on an overlooking point of the city).

By integrating these perspectives, this chapter establishes the theoretical groundwork necessary to critically examine verticality in digital environments. Understanding how space becomes meaningful provides the conceptual basis for exploring how vertical movement transforms spatial experience, deepens presence, and affects player agency.

3.1 Space and Place in Human Experience

3.1.1 Space and Place

What is space? In the novella *Le Petit Prince* by Antoine de Saint-Exupéry (1943), the Little Prince lives on a small planet no wider than a house. It is a narrow space, barely enough for three volcanoes, a rose, and a chair. Yet to the Prince, it is more than a rock in the void, it is his world. Its dimensions are irrelevant. What matters are the rituals that fill it with care: cleaning the volcanoes, tending to the flower, watching sunsets by shifting his chair a few steps. This planet, though minuscule, is imbued with love, routine, and memories. When he leaves this world, the Prince travels through the cosmos, visiting asteroid after asteroid, planets after planets. Each is ruled by an adult trapped in abstraction: a king obsessed with command, a businessman counting stars, a geographer who maps but never explores. These characters reduce space to function, ownership, or quantification. Their planets are larger, but emptier, without meaning. The Prince, though polite, leaves each one unmoved. They are not places he can belong to.

It is only on Earth, vast and overwhelming in its scale, that the Prince begins to understand the nature of attachment. In the desert, he meets the fox, who teaches him the slow, deliberate work of creating bonds. To “tame”, the fox says, is to create ties. Through this process, space changes. A wheat field, once indistinguishable from any other, becomes dear because it now reminds the fox of the Prince's golden hair. A meaningful place is born from emotions. By the end of his journey, the Prince comes to understand that “what is essential is invisible to the eye”. The vastness of space is meaningless without

love, other feelings or some kind of habituation that arose. The desert becomes something more because of the bond it holds. Even the stars change their meaning, for now they laugh like the sound of the Prince's joy. This story, though simple, captures the heart of spatial experience. We create attachments to a location through lived experiences. This lesson offers a compass: that emotions and routine are the architects of meaningful spaces.

Yi-Fu Tuan's work in humanistic geography, particularly in *Space and Place: The Perspective of Experience* (2001) and *Escapism* (1998), offers a phenomenological understanding of how individuals perceive and give meaning to environments. His distinction between *space* and *place* has become an essential concept for interpreting space, including game studies (Calleja, 2011). Tuan (2001, p. 6) defines *space* as openness, freedom, and potential. A realm of possibility that comes before experience. It is abstract and objective. *Place*, by contrast, is created when space is provided with meaning through human interaction. As he puts it: "Place is security, space is freedom: we are attached to one and long for the other" (Tuan, 2001, p. 3). This duality maps neatly onto digital game environments: large, uncharted open-world maps invoke the freedom of space, while hubs, home bases, or other familiar settings offer the comfort and attachment characteristic of place. The author introduces what he calls the "experiential perspective" (Tuan, 2001, chapter 2), a way of understanding space and place as experiences shaped by movement, perception, and time. For Tuan, it is through bodily encounters (walking through a landscape, pausing in a square, revisiting a home) that a person transforms undifferentiated space into place. "Place is pause", he writes. Each pause in movement makes it possible for location to be remembered and valued. When the little prince leaves behind his rose, his volcanoes, and the quietness of his asteroid, he remakes each new world through his personal experience.

So, as it suggests, this transformation from abstract space to meaningful place, is the result of continuous perception and practice. Tuan claims, "what begins as undifferentiated space becomes place as we get to know it better and endow it with value" (2001, p. 6). In gaming, this is mirrored in how players gradually become oriented within a game world. For example in *Star Citizen* (Cloud Imperium Games, 2017), the first time I went to the city of Lorville, my experience was overwhelming and confusing because the place was foreign and complex, but through movement, interaction, and repeated visits, it became familiar, and navigating the city to get supplies became almost second nature.

This process is enriched by the dimension of escapism. In *Escapism* (1998), Tuan discusses the human desire to construct alternate realities, not as avoidance, but as an imaginative engagement with different ways of being in the world. He draws on examples like Disneyland, an environment

meticulously designed to transport visitors into another world (Tuan, 1998, "Fantasies of the Good"). Digital games operate similarly, crafting worlds that players can inhabit, explore, and ultimately make their own. An excellent example of this are the world creations of *Minecraft* (Mojang Studios, 2011) players.



A player's (Andy_JF) Skull Mountain from Reddit https://www.reddit.com/r/Minecraft/comments/9zndqy/skull_mountain_im_building_on_our_server/

Scale and intimacy also affect how space becomes a place. Tuan notes that smaller, bounded environments often foster stronger emotional attachments due to their clarity and coherence. Expansive open worlds like in *The Elder Scrolls V: Skyrim* (Bethesda, 2011) or *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017) encourage a sense of exploration but may complexify the attachment to place until players discover recurring paths to create memories and a routine. Repeated visits to specific areas, such as the Dedsec hideout in *Watch Dogs 2* (Ubisoft, 2016) used as a HQ for operations, reinforce the player's connection to the location. As a result, these locations become centres of memories and player's progress. The emotional resonance of return underscores Tuan's observation that "place is remembered, space is not" (Tuan, 2001).

Building on these phenomenological foundations, the work of philosopher Edward S. Casey offers a crucial perspective that complements Yi-Fu Tuan's distinction between space and place. While Tuan (2001) explains *that* abstract space is transformed into meaningful place through experience and

attachment, Casey's work clarifies *how* this transformation occurs: through the body. Casey argues against a tradition that has privileged time over space, treating space as "the dead, the fixed, the undialectical" (as cited in Matsuda, 2004, p. 260). For Casey, place is not something we merely observe from a distance; it is something we experience through our physical being. He is interested in the "intimate commitment of the subject *within* that landscape" (Matsuda, 2004, p. 262).

This focus on embodiment provides a powerful mechanism for Tuan's ideas. Central to Casey's argument is the concept of "**re-emplacement**"—the idea that when we represent a place, we are not simply creating a copy, but are generating a new context for it through our embodied experience. Casey insists that "the power of place is most fully manifested at the very moment when place and body fuse and lose their separate identities" (Casey, as cited in Matsuda, 2004, p. 271). This concept of embodied emplacement is profoundly relevant to digital games, where players interact with the world through a virtual body—their avatar.

This combined perspective is especially critical for understanding **verticality**. The acts of climbing, jumping, falling, and soaring are intensely bodily experiences, even when mediated through a controller. In these moments, the player's physical intentions and the avatar's actions fuse with the vertical environment. Following the logic of Tuan and Casey, the player is not just traversing an abstract vertical space, but is actively "re-emplacing" themselves within it, transforming it into a meaningful place through embodied action. This forges the deep, kinesthetic connection that is fundamental to the sensation of presence. Their combined work thus provides an essential theoretical link between the physical act of vertical movement and the psychological experience of being in a place.

Building on these phenomenological foundations, the work of philosopher Casey (2002) offers a perspective that complements Yi-Fu Tuan's (2001) distinction between space and place. Casey's work clarifies how this transformation occurs: through the body. For Casey, place is something we experience through our physical being. He is interested in the "intimate commitment of the subject within that landscape" (Casey, 2002). This focus on embodiment supports Tuan's ideas. Central to Casey's argument is the concept of "re-emplacement", the idea that when we represent a place, we are not simply creating a copy, but are generating a new context for it through our embodied experience. He insists that the power of place manifests when the space and the body fuse together to create a unique experience (Casey, 2002). This concept of embodied emplacement is relevant to digital games, where players interact with the world through their avatar.

This perspective is also significant for understanding verticality. The acts of climbing, jumping, falling, or flying are bodily experiences conveyed by a kinesthetic involvement. In these moments, the player's physical intentions and the avatar's actions merge with the vertical layers of the environment. Following the logic of Tuan and Casey, the player is actively "re-emplacing" themselves within it, transforming it into a meaningful place through embodied action. This forges the kinesthetic connection that is fundamental to the Calleja's (2011) model. Thus, this provides an essential theoretical link between the physical act of vertical movement and the psychological experience of being in a place.

Furthermore, Tuan's insights intersect with Henri Lefebvre's concept of *representational spaces* or *lived space* (1991, p. 39). Tuan's focus is primarily experiential and affective, and Lefebvre expands the discussion to include the social and ideological dimensions of space. He describes lived space as emotionally and symbolically charged. The parallels are clear: players engage with digital landscapes not just as navigable geometry (*perceived space*), but as meaningful sites of agency, routine, and belonging (*lived space*). Taking a break in a hot spring in *Hollow Knight* (Team Cherry, 2017) exemplifies this dynamic as it is a great way to ease up and catch your breath from the intensity of the gameplay. It reinforces emotional engagement with the world.

Tuan's ideas help us understand how space becomes meaningful through personal experience and emotional connection. But to go further, we can look at how people actually think, feel, and behave in different environments. This is where environmental psychology comes in. It builds on Tuan's work by focusing on how we form attachments to places through memory, routine, and interaction. While Tuan looks at meaning and experience in a broad sense, environmental psychology gives us more detailed tools to study how places shape our thoughts and actions, and how we shape places in return.

3.1.2 Environmental Psychology and the Sense of Place

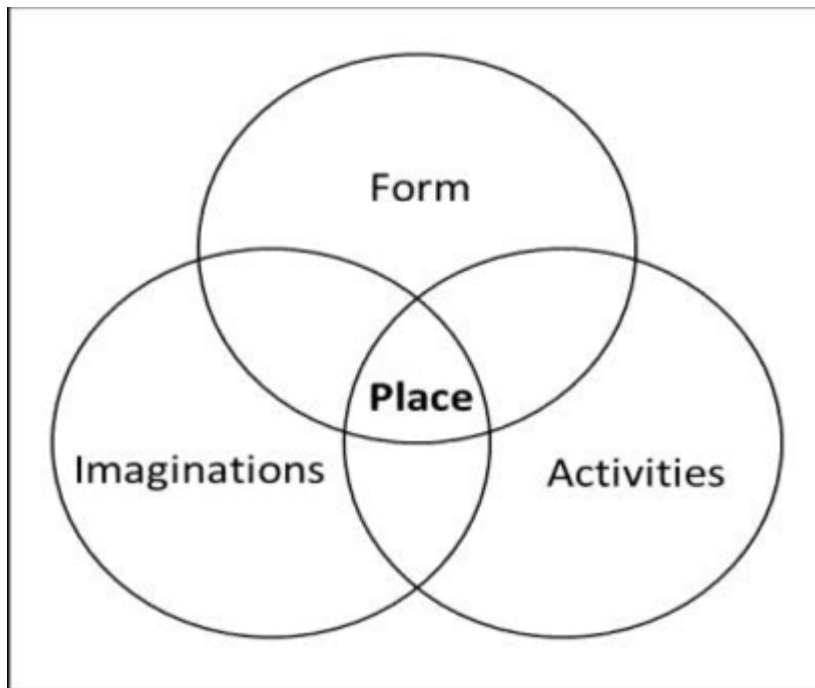
Environmental psychology offers a valuable framework for understanding how people perceive, interpret, and emotionally respond to their surroundings. This branch of psychology explores the relationship between humans and the world in which they act, how the environment around us shapes us as individuals (Ackerman, 2018). Central to this field is the concept of *sense of place*, referring to the cognitive, emotional, and symbolic bonds that individuals form with particular environments (Foote & Azaryahu, 2009). It is also used to highlight the uniqueness and character of a specific location. This concept makes a significant bond with Tuan's idea as it has also been applied to the comfort, safety and well-being constructed by a place. This perspective is particularly fruitful when

applied to digital games, as game designers seek to form meaningful places for the player with spatial layouts, routines, and interactive structures (Bouchoud, 2013).

A sense of place emerges through the interplay of physical characteristics (such as architecture, layout, and detail) and human elements like memory, cultural meaning, and narrative. Scholars such as Edward Relph (Seamon & Sowers, 2008) and David Canter (1977) have emphasised that place is defined by material features, but also by the interpretations and experiences of individuals. They introduce the ideas of place attachment (called *insideness* by Relph) and place identity. Place attachment or insideness, simply put, is the degree of involvement and concern an individual possesses for a particular place. Place identity refers to the uniqueness and characteristics that allows a place to be recognised from others. Tuan (2001, p. 166) exemplifies this idea by writing about French villages who often come with a church at their centre.

From an experiential standpoint, environmental psychology insists that place is observed but more importantly lived. This means that action, movement, and habituation create the emotional bond necessary to get familiar with a location. People come to know a place by inhabiting it, walking its paths, engaging in its activities, forming routines, and memorising. In gaming, players similarly form bonds with digital environments through the same actions. This process transforms the level into a space that talks and might feel familiar and even comforting. There is a relatable feeling in *Minecraft* (Mojang Studios, 2011) when after a long session of mining and putting yourself in danger by facing monsters constantly, you are able to get back to what you call home in-game. You can settle, store the loot and peacefully farm.

David Canter's (1977) model of sense of place asserts that the experience of place is shaped by the triangulation of its physical form, the behaviours it encourages or permits (imagination), and the meanings assigned to it. This is echoed in digital games through the design of behaviour settings: defined, structured areas where specific interactions take place. Behaviour settings in digital games include locations like safe zones (e.g. inside dungeons in *The Legend of Zelda: Phantom Hourglass* (Nintendo, 2007)) or shops/healing areas (e.g. pokémon centres in *Pokémon Diamond* (Game Freak, 2007)) which anchor consistency and logic in the player's mind, which can also be reinforced through a reward system (Swink, 2008, p. 180).



Sense of place model (Canter, 1977)

Relph (Seamon & Sowers, 2008) contributes to this understanding with his distinction between *insideness* and *outsideness*. Insideness describes a deep, affective connection to a place. Relph's idea of insideness is closely related to Tuan's place. He demonstrates that the structure of a place has meaning for human life. A person who feels inside a place will feel safe and enclosed rather than threatened and exposed (Seamon & Sowers, 2008). Conversely, *outsideness* is marked by detachment, alienation, or lack of orientation. In digital games, players often oscillate between these states. Entering a new zone often evokes *outsideness*, but over time, *insideness* emerges, strengthening the player's attachment with the place. Titles like *Hollow Knight* (Team Cherry, 2017), where familiarity with the space is earned gradually through exploring and memorising, exemplify this transformation.

Relph (Seamon & Sowers, 2008) suggests that the greater an individual's insideness with a place, the greater will be their identity with this place (the way individuals define themselves through their relationship with their environment). In some digital games, players actively construct identity by customising avatars, decorating housing, or associating personal stories with particular places. This aligns closely with Calleja's (2011) notion of self-awareness and narrative involvement, where the player's character becomes an extension of self, spatially situated within the game world. Consider *Animal Crossing: New Horizons* (Nintendo, 2020), where players build and personalise islands based on their tastes. Each space becomes a curated extension of the self, rich with meaning.

An important takeaway from environmental psychology is that it cautions against reducing the concept of place to its physical form or geometry. As we have seen, scholars such as Tuan (2001), Relph (Seamon & Sowers, 2008) and Canter (1977) argue that the meaning of a place arises through human perception, intention, and lived experience. Place becomes meaningful through the ways individuals interact with it and the memories they associate with it. This means that a visually stunning game environment can still feel empty or lifeless if it lacks opportunities for meaningful interaction. On the other hand, even low-fidelity graphics in a game can evoke a strong sense of belonging when there is life and space for action. For instance, *Undertale* (Toby Fox, 2015) uses simple, pixelated visuals, yet creates attachment to the place when the player first arrives in Toriel's Home and is given safety, warmth and care to make this place a temporary refuge.

Understanding how people form emotional and cognitive connections to physical environments confirms that spatiality is shaped through interactivity and personal meaning. These insights from environmental psychology are especially useful when thinking about virtual worlds. While game environments are not physically real, players often respond to them as if they were, building familiarity and attachment through exploration and action. This sets the stage for the next section, which explores how digital game spaces become meaningful through player engagement.

3.1.3 Relevance to Game Worlds: Spatial Meaning Through Interaction

The convergence of humanistic geography, environmental psychology, and urban theory reveals that spatial meaning in digital games is more than about visuals, it also emerges through interaction. Drawing from Tuan's (2001) notion of place and Lefebvre's (1991) lived space, it becomes clear that spatial engagement in games is affective and cognitive. Digital game environments' significance, in addition to how they look, also comes from what players do within them. As demonstrated, spatial meaning is produced through movement, action, and repetition. In a digital game, this could be translated to climbing a cliff, fleeing from enemies through alleyways, or returning to a home base after a mission. Over time, the game world becomes a layered terrain, shaped by the player's personal narrative and experience.

This is where Kevin Lynch's (1960) theory of *imageability* provides a functional and perceptual complement to Tuan's and Relph's theories. Lynch identifies five core elements: paths, edges, districts, nodes, and landmarks. Those shape how individuals form mental maps of their environments. These structures improve spatial legibility, facilitating orientation, navigation, and attachment (see section 3.2 for more details on Lynch's theory). When applied to games, Lynch's model offers a toolkit for designers to build their worlds. For example, *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017)

employs highly imageable terrain, where distinct landmarks (mountain peaks, shrines), functional nodes (stables, towers), and intuitive paths foster a sense of navigation and autonomy. Similarly, *Dark Souls* (FromSoftware, 2011) makes extensive use of spatial loops, bonfires as nodes, and dramatic landmarks like the impressive city of Anor Londo or the swampy settlement of Blighttown to help players build spatial memory. Additionally, the game's difficulty increases the emotional weight of returning to hard-won areas. It is worth noting that camera perspective also plays a significant role in shaping how game space is understood and valued (Nitsche, 2008; Nnadi et al., 2008). First-person games offer more immediacy and embodiment with the gameworld. In contrast, isometric or third-person games, such as *Diablo IV* (Blizzard Entertainment, 2023) or *Age of Empires IV* (Xbox Game Studios, 2021), provide strategic distance and a broader overview of the environment (Nitsche, 2008). These shifts in viewpoint influence the kinds of relationships players build with space: embodied and visceral versus reflective and tactical.

Moreover, the idea of procedural familiarity, the repeated performance of movement or action, links digital space with the concepts of routine in environmental psychology. Just as returning to a specific bench in a park can give a sense of comfort, so too does frequently revisiting a particular district in a game world. This relates to the idea of the internalisation of the space in spatial involvement as described by Calleja (2001). For example, in *Counter-Strike: Global Offensive* (Valve, 2012), players navigate more fluidly and effectively as they become more familiar with the map layout. Game designers create opportunities for players to transform unfamiliar spaces into meaningful places. Wirth et al.'s (2007) two-level mode offers further support for this transformation. They argue that players first cognitively construct a spatial model of their environment (e.g., understanding the layout), and then, through affective engagement, perceive themselves as being more part of the world. This dual process reflects the idea of space and place by Tuan (2001), where interaction and perception combine to produce an involving environment.

Ultimately, the transition from space to place in game worlds absorbs the transition from navigability to inhabitation that are the two phases described by Calleja (2011) in his model. Just as Kevin Lynch (1960) associated the concept of imageability with a citizen's ability to navigate, understand, and feel secure within a city, in digital games, high imageability provides a similar sensation. This clarity of spatial structure enables players to move purposefully and recall locations. The integration of Tuan's, Relph's, Canter's and Lynch's theories allows us to see digital environments as fields of action and meaning and also why those worlds matter. Game worlds become meaningful because they are lived. In this section, I explored how space becomes meaningful through experience, emotion, and cultural context. But for players to form these deep connections with game worlds, they first need to perceive

and make sense of the space around them. The next section turns the focus to how players actually see, understand, and move through virtual environments.

3.2 Spatial Perception in Game Environments

In order to dig more deeply into how digital space affects a player, it is relevant to comprehend the process of perceiving space, how those players perceive this digital space. Understanding this requires a multidisciplinary baggage that draws from fields like urban design, cognition and ecological theory. By looking at how people form mental maps, recognise space, and respond to visual cues, we can better understand how digital games foster spatial involvement. As I will demonstrate, spatial perception in digital games extends beyond graphic rendering, it encompasses the construction of internal representations and emotional resonance. Based on Gibson's definition (2015), perception is, in essence, the process by which players interpret the environment, inhabit and engage with the world, and act meaningfully within it (Nitsche, 2008; Calleja, 2011).

A good starting point in understanding spatial perception is Kevin Lynch's *The Image of the City* (1960), which introduced the concept of environmental images: mental maps formed by individuals to navigate and make sense of urban space. As introduced in section 3.1.3, Lynch identifies five core elements that contribute to spatial legibility: paths (routes of movement), edges (boundaries and borders), districts (areas with distinct identity), nodes (key intersections or hubs), and landmarks (easily recognisable reference points). These structures allow individuals to construct coherent mental models of their environments, for better orientation, sense of security, and memory of the space (Lynch, 1960).

Game designers frequently borrow from Lynch's framework to improve spatial clarity and coherence. Titles such as *The Elder Scrolls V: Skyrim* (Bethesda, 2011) utilise a combination of paths (roads and rivers), districts (cities like Whiterun or Riften), and landmarks (mountains, towers) to reinforce this process of cognitive mapping. These environmental signs guide the player's exploration and offer affective attachment points. As Yeleswarapu (2021) notes, highly imageable environments also contribute to a more emotionally engaging and memorable experience.

This model is complemented by Edward Tolman's (1948) research in cognitive psychology, which introduced the idea of cognitive maps. Tolman's experiments with rats in mazes demonstrated that living beings construct internal mental models of their surroundings to navigate effectively. In digital games, cognitive maps allow players to learn terrain, recall locations, anticipate danger, and optimise movement. For instance, players of *Hollow Knight* (Team Cherry, 2017) or *The Legend of Zelda: Breath*

of the Wild (Nintendo, 2017) gradually transition from a foreign feeling to mastering their surroundings as they learn the terrain's structure. This mirrors the psychological shift described by Yi-Fu Tuan (2001), where space transforms into place through interaction and familiarity.

Moreover, Gibson's ecological theory of perception (2015) provides a significant complement to these ideas. Gibson proposed that perception is direct and action-oriented. Individuals perceive their environment in terms of its affordances (the actionable possibilities it offers). He argued that, unlike representationalist theories that pose perception as constructed through mental inference, the environment offers structured information that is perceived as meaningful in relation to the perceiver's body and intent. "We must perceive in order to move, but we must also move in order to perceive" (Gibson, 2015, Chapter 13). In the context of digital games, this means players perceive the game world as actionable. Ladders are climbable, cliffs are glidable, and ledges are reachable, through immediate perceptual cues/affordances designed into the environment.

In this thesis, Gibson's ecological theory of perception, particularly his concept of affordances, is employed as a primary tool for understanding how players interact with vertical game spaces. It is crucial, however, to acknowledge that his theory of "direct perception" is not without its critics. Gibson's work represents a direct and sustained challenge to the cognitivist and representationalist theories that have long dominated psychology. As Costall (1984) explains, these traditional theories postulate an isolated "subject" set apart from its environment, who must then rely on internal mental processes to make sense of supposedly ambiguous sensory data.

Gibson's contribution was to question the very premise of this ambiguity. He argued that perception is not a response to a stimulus, but an act of identifying information that already exists in the environment (Costall, 1984, p. 109, Gibson, 2015). From this perspective, the central question is not how the mind builds a world from fragments, but how the environment itself provides information that specifies the potential for action. This ecological approach is well-suited to the analysis of digital games, and specifically to verticality. Game worlds are intentionally designed systems filled with information to guide player action. The designer's goal is to remove ambiguity and make the potential for vertical action directly perceivable.

Therefore, while cognitivist critiques often focus on impoverished laboratory settings where stimuli do underdetermine their source, such arguments are less relevant to the interactive, and information-filled elements of modern game worlds (Costall, 1984, p. 114). By adopting Gibson's framework, this thesis does not need to speculate on unobservable "underlying" mental structures. Instead, it can focus on a more direct and observable relationship: how the designed information within vertical

game spaces lawfully relates to the player's actions. The concept of affordances thus provides a powerful and pragmatic language for analysing the functional link between vertical design and player agency. In *Mirror's Edge* (DICE, 2008) or *Assassin's Creed IV: Black Flag* (Ubisoft, 2013), the environment is also designed as a functional terrain of actions (walls to run across and climb, ledges to grab, gaps to leap). This would mean that these affordances are directly perceived through interaction. A clear example of this can be seen in *Portal* (Valve, 2007), especially in its opening tutorial levels. The game never tells the player exactly what to do through dialogue or text. Instead, it uses visual cues to teach how the world works. White panels on walls and floors stand out from the rest of the environment and are the only surfaces where portals can be placed. Through simple trial and error, the player learns that these white surfaces afford the placement of a portal. Likewise, seeing a gap or a ledge suggests jumping or shooting a portal to cross it. The environment "shows" the player what actions are possible through design, not instructions. This alignment, where perception informs action, enhances fluency through a more intuitive and immediate spatial understanding (Gibson, 2015; Swink, 2008). Swink's work in *Game Feel* (2008) is particularly significant as it bridges the gap between the more reality based psychology to digital game interaction, arguing that the tactile responsiveness of game mechanics is crucial for creating a strong sense of engagement and agency through embodied control. Consequently, this leads us back to the importance of moment-to-moment control and bodily engagement in Calleja's kinesthetic involvement (2011). Altogether, it makes sense to claim that the intentional outcome, as thought by the game designer(s), of a player's perception of the game world leads to a better involvement towards the digital game and possibly fosters a sensation of presence.



Solving a level in *Portal* (Valve, 2007)

As modern games increasingly emphasise three-dimensional movement, we can see that this depth adds another essential layer to spatial perception (an essential point to consider for verticality). Having different heights in the game world introduces a distinct perceptual logic. It requires players to scan upward and downward, and interpret affordances in that logic such as climbable walls or scalable ledges. Games like *Just Cause 3* (Square Enix, 2015) and *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017) illustrate this quite well because they integrate vertical mechanics (the grappling hook and wingsuit in the first example, climbing and the glider in the second) to expand spatial possibilities, giving players freedom of choice in the game world. These mechanics reinforce player agency by layering spatial meaning and encouraging exploration (Totten, 2014; Yang, 2024).

In game studies, player agency is a core concept that describes the player's capacity to act significantly and observe the consequences of those actions within a game world (Murray, 2017). Janet Murray (2017, p. 126) defines agency as “the satisfying power to take meaningful action and see the results of our decisions and choices”. This idea captures what sets games apart from other forms of media, and it plays a vital role in shaping how players experience and engage with digital games since this influences the feedback loop between the user and the machine. This is where James Gibson’s ecological theory of perception (2015) offers a helpful framework through his concept of affordances to see agency as closely tied to movement and perception. In digital games, affordances,

often communicated visually and physically, become meaningful when understood in the context of what the player can do.

Expanding on this idea, Bódi (2022) argues that agency in digital games emerges from the player's ability to perceive possibilities for action. It is not just about having options but recognising which actions are available and relevant in a given situation. This perception is shaped by how the game world is designed and how clearly it communicates its interactive potential. This relates to the previous *Portal* (Valve, 2007) example. I align this process with Calleja's (2011) theory of player involvement, particularly in relation to spatial and ludic engagement. The sense of presence is reinforced when players feel that their actions have weight and that the environment responds in meaningful ways. *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017) exemplifies this dynamic through its open-world approach. Players are encouraged to explore a vast, layered landscape where nearly every surface is climbable, weather patterns affect mobility (rains make climbing more difficult), and environmental physics offer creative problem-solving opportunities. For example, a player might approach a cliff and, if it's a sunny day, decide to climb it. At the top, the player notices a new shrine, rewarding their decision. After that, they might use a glider to descend and land directly on an enemy camp, with the strategy to engage from above. In each case, the environment becomes an active participant in gameplay, and its vertical structures offer affordances that players learn to perceive and use.

Based on these insights, it is easy to say that agency plays an important role in how players develop spatial understanding. As they move through the game world, they begin to test boundaries, and internalise how the space functions. This mirrors Gibson's idea that perception is shaped by action and the human experience of space. The environment becomes more familiar through repeated engagement, and the player's confidence grows as they sense more control.

An additional idea that can be used to illuminate how players emotionally and perceptually engage with digital space is *psychogeography*, a concept developed by the Situationist International movement. At its core, psychogeography explores the ways in which environments influence mood, behavior, and decision-making (Coverley, 2018). One practice used in psychogeography is Guy Debord's (1958) notion of the *dérive* ("drift"), a form of purposeless wandering through urban landscapes guided by emotional and sensory cues rather than rational planning. It highlights the psychological and stimulating impact of spatiality. In contrast to planned, goal-oriented navigation, the *dérive* prioritises emotional pull and exploring intuitively. This reveals how perception is shaped not only by function but also by the atmosphere of space.

This idea resonates with Gibson's (2015) approach to perception, in which environments offer affordances that are directly perceived and acted upon by the individual. For Debord (1958) and Coverley (2018), the urban landscape affords, in addition to routes and obstacles, feelings of curiosity, alienation, wonder, or unease. In digital games, these affordances become layered with symbolic meaning and interactive feedback, to guide the players into having the emotions and perception as they were designed. Games like *Shadow of the Colossus* (Team Ico, 2005) and *Sable* (Shedworks, 2021) are good examples of this principle. Their expansive, atmospheric landscapes (in the way they create a distinctive mood) and interface design wants the players to "drift" through space without strict objectives, responding instead to environmental cues, scale, and mood.

Shadow of the Colossus (Team Ico, 2005) exemplifies the *dérive* through its vast, largely unpopulated world where traversing this landscape is driven less by explicit objectives and more by atmosphere and landscape. The player often rides for long stretches without combat or dialogue. The user is guided by terrain contours or visual cues such as a distant structure or shaft of light. The deserted environment affords a sense of solitude and maybe even reverence, which aims at encouraging introspection. It produces a slow and gutty rhythm akin to Debord's idea of drifting.

Sable (Shedworks, 2021) offers an even clearer idea of the *dérive*. Set in a desert world, the game strips away combat entirely, foregrounding exploration, puzzle-solving, and narrative fragments. Players navigate across dunes and rock formations, where structures have a pulling aesthetic that stimulates the curiosity of the players. There's no minimap, no imposed path, only a diegetic compass and landmarks that players use to orient themselves. These two games illustrate a link back to Lynch's emphasis on imageability and Gibson's concept of active perception.

The affective mode of engagement that is the drift also connects to Kevin Lynch's (1960) theory of imageability. Lynch argued that spaces become legible and emotionally resonant when they possess distinct paths, edges, nodes, districts, and landmarks. In digital games, these spatial elements are often stylised to elicit emotional response: lonely roads stretching toward unknown ruins, imposing cliffs as psychological barriers, or tranquil districts that suggest safety and reflection. These are clearly represented and used in horror games as they use patterns of obscurity to generate unease, dramatic atmosphere and tension (Niedenthal, 2009). Psychogeography is therefore a notion that explains the effects of affective and aesthetic affordances of the environment through spatial perception on the players. To further support this perspective, neuroscientific studies support the claim that spatial navigation in digital environments engages real-world cognitive systems. Research by Maguire et al. (1997) has shown that the hippocampus (the brain's primary region for spatial memory) is activated during both real and virtual navigation tasks. This implies that digital spatial experiences are not simply

symbolic or imagined but grounded in the same biological processes that govern embodied spatial awareness in the physical world.

Altogether, these points of view reveal spatial perception in digital games as an active and affectively rich process, an understanding that is foundational to this thesis's central argument. This perceptual engagement becomes particularly complex and significant when considering verticality. The introduction of height and depth fundamentally alters the player's experience, demanding more than a two-dimensional cognitive map and introducing a unique set of affordances for action, such as climbing, gliding, and falling. Players perceive these vertical spaces through the symbolic meanings embedded within them. A skyscraper and a cavern carry inherently different emotional and cultural weights, evoking feelings of power, vulnerability, ambition, or dread. This affective charge echoes Henri Lefebvre's (1991) notion that space is never neutral but is actively produced and inscribed with meaning through social relations. In game worlds, verticality is often a primary tool for representing these social structures, such as hierarchy and power. To better understand how these vertical meanings are constructed and how digital environments become sites of cultural expression, we must now turn to Lefebvre's theory of space as socially produced.

3.3 Social Production of Space

In section 3.1, I examined how space acquires meaning through human experience, perception, and emotion. Drawing from Yi-Fu Tuan's (2001) phenomenological approach, I demonstrated how individuals transform abstract space into place through memory, and familiarity. But while Tuan foregrounds the personal and experiential, he also recognises that these experiences are never isolated from culture. Our spatial attachments, routines, and feelings are also shaped by broader symbolic systems (values, stories, and structures of the societies). As Tuan notes, "culture is uniquely developed in human beings. It strongly influences human behaviour and values" (2001, p. 5).

Yi-Fu Tuan (2001) emphasises that while certain aspects of how people perceive and relate to space may be universal, the interpretation and valuation of space are deeply influenced by cultural backgrounds (p. 5). For instance, Tuan notes the significant differences between the Eskimos' sense of space and that of Americans (p. 59). This illustrates how cultural traits inform spatial interpretation. Tuan also recognises that shared cultural traits can go beyond individual experience, it creates common spatial sensibilities within groups, a shared experience of space (p.19). He suggests that societal norms and ideological beliefs influence the design of physical environments and the meaning attributed to particular spaces, such as religious sites or urban areas. This interplay underscores that

space and place are deeply embedded in the societal and cultural fabric that shapes human interaction with their environment.

This is where Henri Lefebvre's (1991) theory of the social production of space becomes an essential theoretical support. Lefebvre shifts the focus from the individual to the collective, to a structural formation. He argues that space is actively produced through social relations, institutional forces, and political ideologies. It is shaped by power, formed by shared values, and contested through use. This perspective enables a better understanding of how spaces reproduce societal values and power structures. In this section, I build on Lefebvre's triadic model of space (1991) to explore how space is socially produced. I unpack the core of this model and its implications for spatial design and then turn explicitly to digital games, to demonstrate how game designers use these notions to build specific digital environments for the players to experience the in-game society through space.

Central to Lefebvre's thesis is the spatial triad, a conceptual model that divides space into three interrelated levels. The first is spatial practice, or perceived space, which refers to the physical and material interactions people have with their surroundings in daily life. This relates to section 3.2 on spatial perception. The second is representations of space, also called conceived space. This dimension involves the abstract and planned aspects of space, often shaped by designers, developers, and world-builders. It encompasses the visual layout, architectural structure, and mechanical logic that define how a game world functions. The third is representational spaces, or lived space, which captures the symbolic, emotional, and experiential qualities of space as it is felt and interpreted by individuals (Lefebvre, 1991, pp. 33–39). In games, this might include a player's personal connection to a specific location, the emotional tone of a setting, or the narrative meaning they assign to their experiences within the environment. The vault 111's platform in *Fallout 4* (Bethesda, 2015) is the location where the player sees the nuclear bombs fall onto Boston and experiences the explosion. This element profoundly marked my experience when I played, because each time I came back to this place, I was living the memories attached to this location again.

For Lefebvre, social space is also a site of power and control. Space is never neutral, it reflects ideological values, cultural assumptions, and dominant systems of production. As he explains: "space serves as a political means of introducing a social and economic structure in such a way that it may gain a foothold and indeed establish its 'base' in a particular locality" (Lefebvre, 1991, p. 151). In addition to being a means of production it is also a means of control, and hence of domination, of power. In digital games, this implies that virtual environments reproduce or challenge dominant ideologies through spatial design. For example in *Watch Dogs: Legion* (Ubisoft, 2020) where the

futuristic city of London features militarised checkpoints with propaganda or districts clearly being controlled by criminal organisations.



A militarised traffic checkpoint in *Watch Dogs: Legion* (Ubisoft, 2020)

This is where Lefebvre criticises what he calls abstract space. A type of space shaped by capitalistic systems of control, order, and economic efficiency. In abstract space, complexity is reduced to uniformity. Spaces are made easy to manage, navigate, and measure in order to appear homogeneous and neutral, often at the cost of individuality, cultural richness, and devoid of specific meaning (Lefebvre, 1991, p. 49). This kind of space is typical of shopping malls, airports, or corporate office buildings found in the western culture. Their layout is designed to guide people in predictable ways. The aim is to discourage loitering, improvisation and therefore personal expression. Everything is clean, controlled, and efficient but also impersonal and often alienating. We can see parallels in digital games. Some level designs prioritise clarity, efficiency, and flow over narrative depth or player creativity. Procedurally generated dungeons in games like *Diablo IV* (Blizzard Entertainment, 2023), for instance, often feel repetitive and interchangeable. These spaces function well for gameplay mechanics but rarely create a sense of place or emotional attachment. On the other hand, this method could be an intentional design to stimulate these feelings to the player. An example is *The Stanley Parable* (Galactic Cafe, 2013). The game unfolds within a series of identical office corridors and sterile meeting rooms, carefully designed to limit player choice and evoke the monotony of corporate life. This environment reflects a system of control, where every path and decision appears preordained. In

this way, *The Stanley Parable* (Galactic Cafe, 2013) dramatises Lefebvre's idea of abstract space, as a medium of ideological reproduction. This is a way for the game to critique how abstract environments can suppress and censor individuality, and spontaneity, by making the player aware of the constraints embedded within the space.

In contrast, Lefebvre proposes the idea of differential space, a space that embraces diversity and critical ideologies (Lefebvre, 1991, pp. 52, 382). Differential space allows people to make it their own, to give it personal or cultural meaning. He points to the vast shanty towns of Latin America, such as barrios and ranchos, as an example of differential space. Despite their poverty and peripheral status, these neighborhoods exhibit a social vitality that surpasses that of more affluent districts or as Lefebvre says, "bourgeois". They organise their space in ways that resist the homogenising forces of authority and urbanisation, and actively defend their unique social and spatial identities in a struggle against other classes. He also claims that spontaneous architecture and planning seen in these neighbourhoods is greatly superior compared to specialists' spatial planning.

Ultimately, it shows that cities are structured environments that reflect the ideologies, histories, and values of the societies that build them. Also, it displays that spatial design has long been a tool for organising social behaviour and enforcing power, which, as Lefebvre demonstrates through his book, also influences cultural expression. The layout of space implies political and cultural intent, as exemplified by the monumental boulevards of Haussmann's Paris, designed to facilitate military control and capitalist expansion, but also its buildings' layout which segregates its inhabitants by class and function (Da Costa Meyer, 2022; Lefebvre, 1991, p. 289). Henri Lefebvre's spatial triad identifies this logic in conceived spaces: the abstract, rationalised vision of planners, architects, and authorities who impose order through ideologies and blueprints. Yet this vision is always contested by lived space, where people reinterpret, resist, or adapt spatial expectations. Let's imagine: public squares become protest grounds, alleyways host informal businesses, staircases double as social gathering points. In this sense, space is not only produced from the top down, but is also redefined by the citizens from the ground up.

In games, differential space can be seen in titles like *Disco Elysium* (ZA/UM, 2019), where players navigate the city of Revachol as a textured, political world full of contradiction and history. Players interpret its meaning differently depending on their choices, values, and character development. This invites reflection and multiple interpretations, closer to Lefebvre's ideal of lived and meaningful environments. These dimensions of space (conceived, perceived, and lived) describe the sociopolitical textures of real environments but also echoes in the constructed spaces of digital games. The following

section explores how spatial design in games embodies this three-points logic, which functions as a cultural signifier, and structures narrative.

Throughout this section, we have seen how space in digital games plays an active role in shaping experience. By looking at how environments are designed, how players move through them, and how meaning is created over time, we gain a better understanding of how games communicate through space. Using theories from scholars like Tuan (2001), Lynch (1960) and Gibson (2015) as well as frameworks such as Lefebvre's (1991) spatial triad, I explored how spatial design in games involves complex reactions from players and methods from designers to create a specific and meaningful experience. This discussion builds on the ideas from previous chapters. In Chapter 2, I looked at presence and player involvement; in Chapter 3, I examined how space and perception influence the way players engage with game environments. Now, with these ideas in place, I am ready to focus on the main subject of this thesis: Verticality. This phenomenon adds a new layer to how space is understood and experienced. It affects how players move, how environments are structured, and players' experience with the digital game. In the next chapter, I will look deeper in how vertical space navigation impacts gameplay, storytelling, and spatial involvement, and consequently, how it contributes to the broader sense of player presence in digital games.

Chapter 4: Verticality in Game Design

In the biblical story of the Tower of Babel, humanity builds a towering structure in an effort to reach the heavens and claim divine-like power. This act of vertical ascend by ambition becomes a symbol of aspiration and pride. Yet the result is confusion and discord. God scatters the people around the world and confounds their languages. This halts the construction of the tower and dissolves humanity's unity. The tower thus becomes an enduring symbol of reaching upward, but also of the tension between control, ambition and chaos.

This ancient narrative offers an interesting image through which to view verticality in game design. Just as the builders of Babel sought meaning and power with their vertical construction, players in some modern open-world digital games often ascend towers, climb mountains, or descend into dungeons in search of quests and treasures, or enhancing their skills. Vertical movement transforms the player's relationship with space since it adds a psychological and symbolic layer to navigation. After all, personal development has the habit of using vertical perspective to transfer emotions and ideas. You climb a mountain or build a tower to find enlightenment. You descend from a plateau to start an introspection, or rather, you lay the foundations of your construction to reflect humility. This is illustrated in the methods used by *Fabulous* (2018) for example, a daily habit tracker app designed to improve mental health through habits fueled by behavioural science, and their visual journey used to help the user progress features mountain ascends and descends into volcanic pits as to visualise illumination and the demons that will try to discourage you from your healthy habits.

Yet the story of Babel also reminds us that verticality carries risks. Ambition can overreach. The higher we climb, the more dangerous the fall. In digital games, moving vertically often introduces a balance between reward and danger. High vantage points can offer strategic advantage or inspiring views. However, they may also expose the player to threat, disorientation, or failure. For example, a miscalculation of Link's stamina when climbing a cliff in *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017), or a leap of faith (jumping from a very high point) in *Assassin's Creed II* (Ubisoft, 2009) is only safe with precise inputs. Vertical spaces require skill, thinking, and knowing the environment.

As we will also see in this chapter, vertical structures in digital games carry cultural, emotional, and functional weight. Towers, cliffs, and caverns shape how players move, perceive, and feel, but also the mechanics that allow players to move inside this space. In *Watch Dogs: Legion* (Ubisoft, 2020), using a drone to take some altitude allows the player to survey the area and identify a strategy to clear a

level. If the player chose to play the level frontally (identify the enemies only on an horizontal plane and use their weapon spontaneously), the sensation and goals would be very different. The Tower of Babel also reminds us that verticality is rarely neutral. It can signify domination or hierarchy (see section 3.3.2). In game worlds, vertical structures may embody social inequality, limit access, or reflect ideological divides. As discussed in Chapter 3, space is socially produced, and spatial design often encodes specific cultural values. Columbia (a floating city in *BioShock Infinite* (2K, 2013)) is high above the earth, and as a white-supremacist regime, aims to reflect a worldview that sees itself as superior, untouchable, and isolated.

This chapter examines verticality as a spatial, symbolic, and interactive dimension of game design. Building on earlier discussions of presence (Chapter 2), the meaning of space and perception (Chapter 3), I explore how verticality reshapes the player's sense of presence, and overall engagement. I will demonstrate that vertical design invites players to experience the game world differently. It layers movement with significant meaning, and improves player involvement. While the story of the tower of Babel ends in disunity, in digital games, verticality prompts players to reconsider their place in the game world and generate emotions.

4.1 Defining Verticality

In this thesis, verticality refers to the structuring and organisation of space along the Y-axis. That is, the vertical dimension of movement and spatial arrangement within a game environment. Unlike horizontal design, which primarily emphasises navigation across flat or ground-level planes (the X and Z axes), verticality introduces an additional spatial layer that involves height, depth, and elevation. This can manifest through features such as cliffs, towers, multilevel architecture, staircases, floating platforms, underground passages, aerial environments and more. Verticality is also the intentional design of game spaces to incorporate vertical routes and spatial compositions that have multiple layers. It is present in how levels are constructed and how players access or view parts of the world. This means that vertical space can be represented statically (tall buildings, chasms...) or dynamically, through the player's ability to move vertically by, for example, climbing, jumping, gliding, flying, falling, grappling.

Importantly, verticality is a concept that is both visual and architectural. It affects the geometry of the world, the layout of environments, and the relationship between different levels of space. In level design, to stimulate this vertical feeling and make it intuitive to the player, the idea is to generate a "downward flow" and "upward flow" with structures that stack, ascend, or descend, creating layers that can be explored (Yang, 2024). Verticality can be embedded within specific level design (e.g., multi-

floor buildings), open-world layouts (e.g., mountain regions), or mechanics (e.g., vertical combat or puzzle-solving).

Historically, verticality has been present in game design since the early arcade era with games like *Donkey Kong* (Nintendo, 1981) where the player needs to ascend a series of platforms to rescue a princess, or *Ice Climber* (Nintendo, 1985) in which the player climbs mountains to retrieve stolen vegetables from the antagonist. Although these digital games were in two dimensions (2D) and were linear in their approach to verticality (go up and down), they introduced basic ideas of vertical progression with obstacles. As game engines advanced to three dimensions (3D) environments, verticality also became more sophisticated. Players climb massive machines in *Horizon: Zero Dawn* (Sony Interactive Entertainment, 2017), climb cliffs and towers in *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017), and glide from high vantage points with wingsuits in *Just Cause 3* (Square Enix, 2015). Regarding verticality, 3D games also provide players with more complex camera controls and a deep complexity to how the environment can be navigated which influences the player's perspective and expands gameplay possibilities.

The next step to fully understand the impact of verticality in digital games, is to examine how player perception and interaction applies to vertical spaces. Because verticality is not just about height or structure, as we will see, it is about what the environment invites the player to do (climbing a wall, jumping across a gap, etc...). These affordances, as explained by Gibson's (2015) ecological theory of perception and unfolded in section 3.2, help understand how players interpret and respond to their surroundings. The following section explores how vertical game design supports meaningful movement, decision making and engagement.

4.2 Verticality and Affordances

Verticality is fundamentally tied to how players perceive and act within virtual environments. This can be understood through Gibson's (2015) ecological theory of perception, particularly his notion of affordances (the action possibilities offered by an environment to a perceiving agent), making them relational. A ledge affords climbing because the player has the ability to climb and it takes back to real-life references. This means that cliffs, rooftops, ladders and canyons invite players to act upon it. Even if those spatial designs were purely decorative, during the early stage of the game when the player learns, there is a high chance for them to try, because there is a link between previous game knowledge and real-life experience that influences this perception. They shape how players move, strategise, and experience the world by suggesting what is possible.

As illustrated in section 3.2, Gibson's theory aligns closely with how player agency is conceptualised in game studies. As Bodi (2020) argues, agency in digital games emerges from the player's ability to enact meaningful change within a system. Verticality adds a critical dimension to this relationship, because it expands the spectrum of affordances and deepens the range of player choices like strategic options and more possibilities to approach a challenge (Grey, 2021). For example, in *Mirror's Edge* (DICE, 2008), rooftops, walls, pipes, and railings are interpreted by players as tools for movement. The game's first-person perspective influences this embodied perception: the feeling comes from momentum, timing, and flow of movement. This responsiveness stimulates a strong sense of agency, as affordances linked to verticality create a fluid relationship and a playground for improvisation. This significantly shapes the experience of a player, making the environment feel more tangible. Verticality, in this sense, becomes a dynamic medium for expressing intentionality and skill.

Kevin Lynch's (1960) concept of imageability (the ease with which a space can be understood and mentally mapped) offers valuable insight into how players perceive and navigate vertical forms in environments. While Lynch originally focused on urban landscapes, I have illustrated in Chapter 3 how his five elements (paths, edges, districts, nodes, and landmarks) apply meaningfully to digital game spaces. But in an additional dimension, verticality complicates and enriches these elements. A path can include a vertical climb or descent, landmarks can be suspended towers, districts can feature different kinds of elevations and nodes can appear across different altitudes, very similar to how it is structured in *The Legend of Zelda: Tears of the Kingdom* (Nintendo, 2023). In this context, verticality serves both navigational and experiential functions, encouraging players to internalise height as an essential part of orientation and to make decisions. Consequently, vertical structures, since they anchor cognitively, enhance spatial memory by improving spatial imageability. All in one, this means that verticality expands player agency and emergent behaviour as well as provides support for engaging exploration for players.

In order to provide further discernment and a more practical demonstration of these theories on digital games, I will conduct a concise but insightful case study of a specific digital game. This will act as a deeper exemplification of my application of the theories on the verticality effect. As argued by Fernández-Vara (2019) Game analysis is a crucial tool for understanding how games function as both interactive systems and cultural artifacts. It allows us to explore their mechanics, narratives, and player experiences, revealing deeper insights about their impact on society and their artistic significance. By dissecting a game's context, design choices, and reception, I can uncover how it shapes player interactions and how it brings meaning.

Focus: The Legend of Zelda: Tears of the Kingdom

In TotK, short for *The Legend of Zelda: Tears of the Kingdom* (Nintendo, 2023), verticality plays a central role in shaping how players perceive and interact with the game world. This digital game case study illustrates how the game's vertical design exemplifies Gibson's theory of affordances, extending the player's agency through creative exploration and experimentation with space. I have chosen this game since it features modern level design with a significant focus on vertical structures. Also, TotK has been widely praised for its greatly refined exploration and open-world setting further improving the already appreciated environment of the precedent game in the series (Marks, 2023).

The game is set in Hyrule, a vast open world composed of three primary layers: surface, sky, and underground. Which presents original vertical configurations. High peaks, plains, floating islands, and caverns redefine how players approach exploration. As world decorations, they work well, but these spatial formations are also designed to invite specific forms of movement available to the player: climbing, gliding, freefalling, and building aerial vehicles using the new Ultrahand system. The player is given a network of actionable possibilities.

Using James Gibson's theory of perception based on affordances (2015), we can understand how the game communicates possibilities for action through its design. Surfaces and environmental features offer clues about what players can do. Cliffs can be climbed, floating islands can be glided toward, and certain ceilings can be traversed using the Ascend ability. Instead of being introduced to long tutorials, players discover them through trial, perception, and experimentation. This fits with Gibson's idea that affordances are directly perceived through direct engagement with space.

This design approach supports a strong sense of agency, as described by Bodi (2022). Agency in games emerges when players can form intentions, act on them, and see meaningful results. In TotK, vertical movement is almost infinitely extendable. Players build flying machines, manipulate or merge objects with abilities, or combine tools to reach less accessible areas. These tools allow for creativity and choice, which, as a result, players feel that they are in control of the outcomes and directing their in-game experience.



The large elevation difference in ToTK's landscape

In TotK, verticality is also used to support navigation and orientation. Kevin Lynch's (1960) concept of imageability helps explain this. Large towers, floating structures, and distinct shrines serve as visual landmarks like Skyview Towers. As discussed in chapter 3, these points help players form mental maps of the world and guide exploration, because it enhances imageability. Vertical features act as both destinations and reference points. They offer direction while adding meaning to the terrain. Edges, such as the horizon between land and sky or the descent into the Depths (an underground map), act as powerful perceptual districts that organise the world into distinct but connected realms.

TotK shows how vertical space can expand what players are able to do and how they think about movement. The game gives players the tools to interact with their environment in meaningful and flexible ways. By combining mechanical systems with a clear visual design, it creates an environment where players interpret, act, and respond to space. It applies theories of perception, agency, and spatial legibility in a way that enhances both gameplay and engagement. This makes it a strong example of how vertical design can shape the experience of being in a game world by extending affordances possibilities.

Having explored how vertical environments in games provide actionable affordances and shape player agency through perception and interaction, it is important to consider how the player's reactions are felt through the body. The next section turns to the concept of game feel, focusing on the sensory and

kinesthetic dimension of vertical movement. By examining how mechanics such as climbing, falling, jumping are rendered through control responsiveness, audiovisual feedback, and player embodiment, we can better understand how verticality contributes to the engaging and emotional texture of gameplay.

4.3 Verticality and Game Feel

Verticality in digital games is also a question of how movement through vertical space feels to the player. Thus I focus on the concept of game feel as introduced by Steve Swink (2009). It refers to the sensory, mechanical, and emotional feedback loop that occurs when a player interacts with a game system. It encompasses how responsive the controls are, how movement is communicated through animation and audio, and how intuitive the system becomes through repeated use, which is also a concept discussed by Calleja (2011), he calls incorporation.

According to Swink, game feel is “the tactile, kinesthetic sense of manipulating a virtual object”. Swink explains that game feel is a quality that emerges from three building blocks (2009, p. 2-5). First, real-time control, as a specific form of interactivity (between the user and the computer), is the ability for players to directly manipulate virtual objects in a responsive way. Second is simulated space which is explained as the physical interactions within a game world, including collision detection and level design. The third building block for game feel is polish. It refers to the visual, audio and animation effects that enhance interactions without altering the core simulation (e.g. dust particles at a character’s feet when sliding, a crashing sound of two cars colliding, a camera shake to highlight an impact). Based on these building blocks, game feel determines how satisfying it is to perform actions, and movements like jumping or climbing. Combined they aim to create a sensation of control and appeal in digital games. These elements become especially interesting when looking at verticality since vertical design affords a great number of actions and possibilities, which influences the game feel.

Investigating movement also connects to Calleja’s (2011) notion of kinesthetic involvement, where the player is engaged through physical interaction with the game world, as seen in Chapter 2. It emphasises that movement systems, and the bodily effort they simulate, are central to his model of player involvement. Which leads us to vertical mechanics, such as gliding or grappling, that extend this bodily engagement. Here, I explore how vertical movement contributes to game feel, and how verticality can amplify kinesthetic responses through design, which, in turn, influences strategy, decision-making and emotional response.

This means that verticality in digital games is felt. It shapes the player's experience through movement, difference in rhythm, and embodied engagement. The act of ascending or descending introduces rather unique physical sensations. This kinesthetic quality of vertical movement (how it feels to move) relies on precise control, audiovisual feedback, and the responsiveness of the game's mechanics. As I highlighted earlier in the section, Swink (2009) defines game feel as the moment-to-moment experience of controlling a game, as a foundation for the player's tactile and sensory connection with the virtual world. Vertical movement sharpens this connection by requiring specific gestures, like holding a jump button for longer airtime or timing a grappling shot while free falling. When done well, these actions feel natural, almost instinctive, which Calleja (2011) emphasises as important aspects of a player's kinesthetic involvement in a digital game.

Since Calleja's (2011) notion of kinesthetic involvement describes how players become engaged through their physical interaction with game systems, verticality deepens this engagement by introducing more complex mechanical interactions in the space that keeps the player connected to the virtual world. Climbing in *Assassin's Creed II* (Ubisoft, 2009) demands the player to choose their path up carefully. Jumping from the climbed tower to perform a leap of faith asks the player to mindfully choose their position because otherwise it could fail and result in the avatar's death. In *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017), gliding or falling introduces a sensory contrast between moments of peace allowing to check the environment with the glider or the risk of crashing on the ground. This alters how players emotionally engage with their surroundings. Adding to that, the nice feeling of gliding is limited because it drains stamina, which threatens an imminent fall in the void if the player drains it all. This means that vertical spaces often challenge the player's sense of balance, control, and flow. If the player chooses to climb a wall in the middle of battle, it will slow the pace and significantly raise tension because of vulnerability. These contrasting sensations stimulate the player emotionally and, as Chapter 3 illustrated, amplify their connection to the environment. This highlights the effects of verticality on the kinesthetic quality because it shapes the tempo of interaction and brings rhythm to space that players learn to perform with their bodily actions and attention. This is a crucial part of how verticality deepens player involvement.

Considering that verticality affects how players move, think and interpret, verticality (from Calleja's (2011) ludic involvement perspective) also introduces layers of strategy and decision-making for players. In real world scenarios, controlling the high ground provides a tactical advantage, and this has been reiterated in books and films (e.g. *The Art of War* (Sun Tzu, 400 B.C.E.) and *Star Wars: Episode III - Revenge of the Sith* (Lucas, 2005)) and this is also significantly present in digital games that feature 3D environmental combat. In addition to vertical obstacles asking players to sequence actions

carefully, manage stamina, or master movement systems, positioning in the vertical space becomes a critical part of play, where being elevated offers opportunities for ambush, defense, or escape. This kind of spatial reasoning enriches decision-making. This dimension of vertical play aligns with Calleja's (2011, p.41) discussion around forms of engagement such as gameplay being more than direct inputs, because they also involve the mental efforts required to construct a strategy. This is where players plan, seek efficiency, and engage with problem-solving, and verticality enriches the process as it adds new variables to consider which directly engage with the player's ludic cognition (tactical decision-making, risk-reward assessment and navigating to game objectives).

All in one, Steve Swink's (2009) concept of game feel and Gordon Calleja's (2011) notion of kinesthetic involvement, one of the six dimensions of player engagement, help us understand one of the dimensions where verticality reinforces presence. The tactile response to player input makes spatial movement more involving and embodied. The physicality of a jump or tension of a climb creates a feedback loop that draws attention to the player's movement in space. As a result, verticality becomes a sensory experience that stimulates the player to participate in the virtual space. This reinforces player presence as actions are felt through friction which challenges the player and links to the intuitive sensations of an individual by connecting to real-world sensations.

Building on this, Calleja (2011) introduces the concept of incorporation, which describes a sensation of presence in digital games, where the player, by mastering the control of an avatar and inhabiting the virtual environment, comes to feel that they are inside the game world itself. Vertical mechanics contribute to incorporation by increasing the sense of embodiment and spatial awareness. When players traverse environments, verticality supports them in internalising the avatar's capabilities. Since these movements require a heightened awareness of spatial relationships, it encourages the player to mentally and physically "inhabit" the avatar. In this dimension, we can say that verticality is a mediator of presence.

This embodied engagement also speaks to Yi-Fu Tuan's (2001) perspective on space and place. As presented in Chapter 3, Tuan explains that space becomes a place through experience and movement (through repeatedly acting, moving and identifying meaning in the space). Verticality deepens this shift. Climbing, jumping or exploring a deep cavern are navigation acts that are meaningful encounters with the environment. As players continue to move through these locations, their memories and emotions construct a relationship with the terrain itself. This process from space to place also offers a first foundation for understanding verticality as a potential narrative device. From Tuan's viewpoint, moving through the game world builds spatial meaning for the player which is an important aspect of player engagement.

Focus: The Finals

I have conducted a brief case study of a digital game that will help me exemplify the claims made in this section. Similar to section 4.2, this game analysis provides useful practical insights on applying verticality featured in this digital game to the notions of bodily actions and kinesthetic involvement. *The Finals* (Embark Studios, 2023) is a team-based first-person shooter that merges highly destructible environments with fast-paced combat and vertically layered environments. The main goal is to retrieve an objective to bring it to a specific location in the arena and defend it against other teams trying to steal it. While it stands out for its great visuals and technical innovation (Koreis, 2023), what truly defines the player experience in this game is its deep reliance on verticality as a central pillar of game feel, spatial understanding, and presence. The game also features mobility-based classes (Light, Medium, Heavy) where each one affects the ability to move in the virtual environment (e.g. Light runs faster than Medium and Heavy).

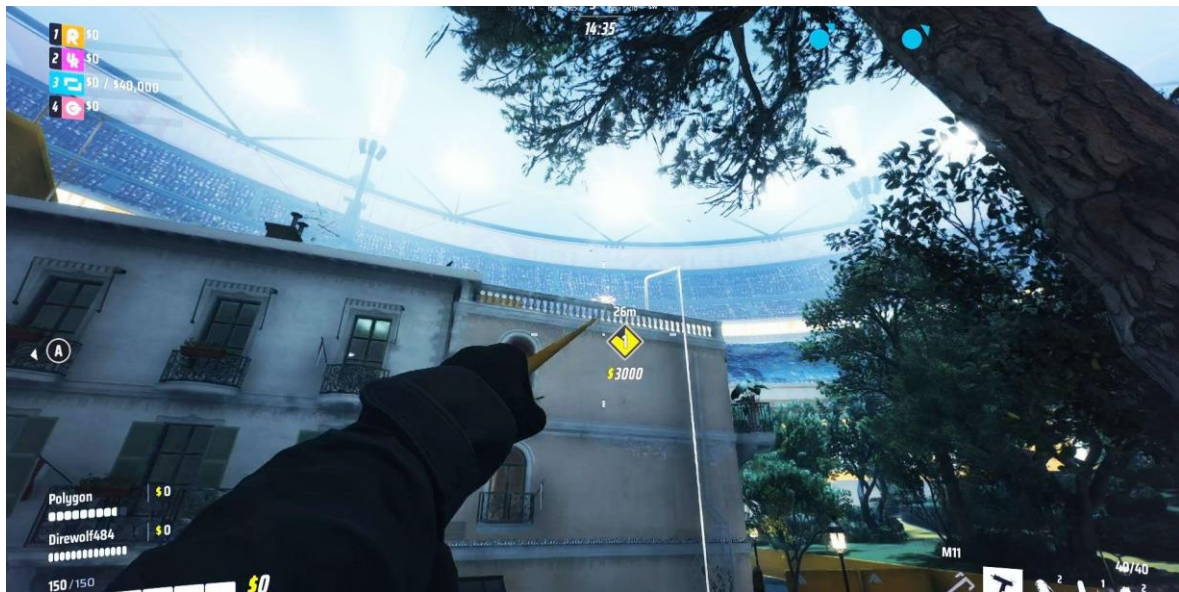
At its core, *The Finals* challenges players to navigate and fight across multilevel urban environments composed of skyscrapers, rooftops, stairwells, elevators, jump pads (pads that launch players in the sky) and zip-lines. The star piece of the game is that these combat arenas are constantly changing, often shaped by player action. A five-storey building can be completely demolished as a strategic decision (e.g. getting easier access to the objective and surprising the defenders). This dynamic combined with vertical structures makes combat feel situational and improvisational. This encourages the player to build reflexes, master the avatar's movements and also come up with creative strategies.

One clear example of embodied engagement can be found in how *The Finals* encourages players to move fluidly across multiple elevations using tools like zip-lines, jump pads, or even classes such as the Light character, who can “dash” and use this ability as a double-jump and sprint through the air. These vertical movements influence the feeling of movement, strategy, and team coordination. For example, having the high ground and dropping through a ceiling to flank enemies are tactical decisions affecting player's ludic engagement and produce immediate, bodily sensations that enhance kinesthetic involvement.

This also echoes Steve Swink's (2009) idea of game feel, where the difference between sprinting across a rooftop and crashing through a window into an enemy's defense generates a sense of gravity, friction, and inertia. This makes vertical traversal feel weighty and expressive.

Considering Calleja's (2011) concept of incorporation, in *The Finals*, players must continuously internalise their avatar's bodily capabilities. Such as how far they can grapple, or how much damage

they deal when landing on the enemy with the Heavy class ability. These are learned through practice and conscious play. For example, a Light class player may get to the top of a building using a grappling hook to coordinate a surprise attack from above. The success of this action depends on how well the player has internalised the timing and reach of the tool, along with the spatial layout of the building. As such, the vertical space becomes fully incorporated into the player's cognitive and bodily understanding of the game world.



Using the grappling hook in The Finals (Embark Studios, 2023).

The destructibility of the environment brings an interesting facet that further enhances the relationship between verticality and presence. Players are not limited to roads, staircases or already-placed ziplines. They can bring down entire structures to reshape the level's elevation. A recurrent strategy from the players is, for example, blowing out the floor under the objective to bring it down and have better chances at stealing it. It gives players a strong sense of authorship over space. Interestingly, this ability to alter the environment reinforces the player's sense of agency discussed in the previous section.

The interplay of verticality, mobility, and spatial design in The Finals emphasises the transformation of space into place, in line with Yi-Fu Tuan's (2001) theory. The cityscape of the arenas in the game becomes meaningful through movement and repeated interaction. For example, I learned to use the wooden ceilings in the Kyoto map as hidden spots to ambush enemies or hide from them during a defending phase.

Hence, *The Finals* provides an insightful case of how verticality is formed into game design to enhance game feel, spatial meaning and therefore, reinforce the sensation of presence. It is a game that uses verticality to invite players to feel and think about the virtual environment. It also demonstrated that verticality carries expressive and symbolic value. In the next section, I examine how vertical space can function as a narrative device, telling stories, reinforcing themes, and enriching the player's narrative involvement.

4.4 Verticality as Narrative Device

The question of how narrative functions within digital games has long been a central and often contested topic in game studies. The early 2000s saw a significant debate between ludologists, who emphasised the procedural and rule-based nature of games as fundamentally distinct from narrative forms (digital games tell stories) and narratologists, who approached games as narratives themselves (Aarseth, 2012; Juul, 2001). This dichotomy has since been nuanced by scholars who acknowledge that games may simultaneously support narrative structures and emergent gameplay systems (Jenkins, 2004).

Gordon Calleja (2011), for his player involvement model, argues that narrative in games should not be confined to cinematic or literary conventions but understood through the player's experiential engagement with the game world. For Calleja, narrative is a co-created process, shaped by the interplay between designer-authored content and player-driven actions. Rather than positioning narrative and gameplay in opposition, he reframes them as interdependent systems: games “tell” through interaction, environment, and choice as much as through cutscenes or dialogue. Here we notice that space plays a role in generating stories, and it brings us back to our theorisation of space with Tuan (2001), Casey (2002) and Lefebvre (1991) who all argued that space is dynamic and lived through human experience which constructs meaningful memories and consequently stories.

4.4.1 Narrative Production of Space

To support this, urban theorists like Kevin Lynch (1960) and Jane Jacobs (1961) recognise the narrative and social functions of spatial design. As we have seen in Chapter 3, for Lynch, elements like landmarks and paths help individuals form mental maps that support orientation and personal interpretation. Jacobs, meanwhile, emphasises the “eyes on the street”, also called natural surveillance. In short, through environmental design, natural surveillance is the organic safety and vibrancy created by encouraging mixed-use neighborhoods, maximising visibility by removing lurking places and adding street lights, and the presence of people for more social interaction in order to limit

the opportunity for crime. Both vision resists abstract, homogeneous urban planning, arguing that modernist urban planning overlooked and oversimplified the complexity of human lives in diverse communities. Instead supporting environments that prioritise human meaning and the needs of residents. This clash of ideologies illustrates that spatial design, then, is also social and cultural filled with stories. Monuments speak of national pride or trauma. Borders define who belongs and who doesn't. Religious architecture shares its interpretation of the world.

Just as cities reflect social values, so do virtual worlds, as forwarded by Vella & Bonello Rutter Giappone (2018), because game spaces are authored environments with expectations for perception and interaction. They note that cities in role-playing games (RPGs) like *The Witcher III* (CD Projekt, 2015) and *The Elder Scrolls V: Skyrim* (Bethesda, 2011) often have ideological underpinnings linked to the Middle Ages and capitalist ideology. Reflecting Lefebvre's (1991) spatial theory, we can understand game environments as constructions that, in addition to providing gameplay affordances, communicate cultural values and narrative meaning. For instance, the imposing vertical skyscrapers with luxurious interiors in *Cyberpunk 2077* (CD Projekt, 2020) visually represent hierarchical oppression when compared to its overcrowded, crumbling districts below. In *Ghost of Tsushima* (Sucker Punch Productions, 2020), sacred Shinto shrines and torii gates add functionality to the gameplay by providing fast travel points, but they also reinforce themes of honour, purity, and spirituality. Likewise, some of Ubisoft's games reconstruct culturally specific cities, like Renaissance Florence in *Assassin's Creed II* (Ubisoft, 2009) or Ancient Greece in *Assassin's Creed: Odyssey* (2018), also as living expressions of sociopolitical tension and heritage (Vella & Bonello Rutter Giappone, 2018).

Consequently, this makes space an active narrative mechanism. The progression through the game world, the spatial arrangement of key elements, and even the affordances present in the environment can communicate stories without a dialogue. Vella and Bonello Rutter Giappone (2018) argue that game cities function as centres organising the surrounding landscape while guiding the player's spatial and phenomenological experience through their avatar. This embodiment allows players to interact with the city's structures such as its boundaries, pathways, and possible encounters, each contributing to a progression in the digital game's story. The opposition between inside and outside, reinforced through gated access and walled architecture, serves as a visualisation of exclusion. Often, players are required to earn entry, reinforcing themes of privilege and restriction. Moreover, traffic (pedestrian or vehicular) within game cities reflects their role as hubs of economic, political, and social activity. This mirrors real-world urban dynamics. By structuring the player's interactions within game space, these design choices enable dynamic storytelling that situates the avatar within a broader cultural and

historical framework. This reinforces spatial narratives through lived experience. One example Vella and Giappone (2018) use to illustrate the narrative role of game space is the city of Novigrad in *The Witcher 3: Wild Hunt* (CD Projekt, 2015). They discuss how its spatial design positions Novigrad as a central hub within the game world. The city acts as a symbolic and functional space, with economic and political hierarchies that the player experiences. Verticality here projects a narrative to the player with Novigrad's gated access and divisions between affluent and impoverished districts. These differences in elevations serve as spatial markers of privilege and exclusion, which affects the player's progression and situates the player as an outsider.



The medieval city of Novigrad that requires earned entry (official papers) in *The Witcher 3: Wild Hunt* (CD Projekt, 2015)

Finally, spaces are functional in how they structure player interaction and agency. As I highlighted in section 3.2, game designers rely on architecture, lighting, and environmental cues to subtly direct attention, create tension, or pace a player's progression. Totten (2019) outlines how principles of real-world architecture, such as focal points, symmetry, and spatial rhythm, are essential tools in level design. In *The Legend of Zelda: Tears of the Kingdom* (Nintendo, 2023), the modular vertical design encourages exploration through climbing, gliding, and surfing. Players can approach objectives creatively and physically engage with space in meaningful ways.

These three dimensions (cultural, narrative, and functional) operate in harmony. Cultural ideologies shape narrative spaces by giving meaning to structures and this narrative weight is distributed through

functional design. Similar to *Dishonored* (Arkane Studios, 2012)'s city of Dunwall communicates themes of plague, class oppression, and corruption through visual semiotics and gameplay loops (a repeatable sequence of actions the players engage in that keeps them playing over and over again) such as stealth and surveillance.

This practice for digital games to turn spatial design into narrative experience has a name: environmental storytelling. It involves embedding narrative elements directly into the space itself (through layout, visual cues, architecture, sound, interactivity, and symbols). It is a method by which players encounter stories as active participants, because it necessitates these players to explore the environment consciously in order to construct meaning and interpret the story behind their surroundings. This narrative capacity of space has been widely recognised in both game studies and design practice (e.g., Bouchoud, 2013; Carson, 2000; Jenkins, 2004). It also emphasises Henri Lefebvre's spatial triad that links real environments and virtual ones. His model of conceived space (the designer's plans and logic), perceived space (the space as traversed and inhabited), and lived space (the symbolic and emotional interpretation by users) is helpful to understand how environmental storytelling operates across multiple layers of game experience.

Looking at *The Last of Us* (Naughty Dog, 2013). Its post-apocalyptic urban landscapes are also used as narrative artifacts. Crumbling skyscrapers, overgrown parks, graffiti, and found notes narrate the collapse of society and the presence of the last survivors of humanity. In the hospital sequence, the spatial layout features isolated corridors, limited sightlines, ambient silence, which aligns the player's emotional tension with the protagonist's ethical dilemma. Here, all three spatial dimensions of Lefebvre's model converge: the designer's layout (conceived), the player's infiltrating the space (perceived), and the moral-emotional weight of that moment (lived).



A survivor camp in *The Last of Us* (Naughty Dog, 2013)

Henry Jenkins' essay *Game Design as Narrative Architecture* (2004) provides a comprehensive framework that explains how space in digital games can specifically act as a narrative medium. Jenkins outlines four primary types of environmental storytelling: evocative spaces, enacted narratives, embedded narratives, and emergent narratives.

Evocative spaces are game environments that connect to stories, symbols, or memories players already know from media, history, or (game) culture. These spaces rely on players to recognise familiar elements in order to add emotions and signification. These environments feel rich and alive because they remind players of something bigger. They use style, architecture, or visual themes to suggest a backstory, much like how a film set can suggest a world beyond the camera. The city of Rapture in *BioShock* (2K, 2007) is a good example. Inspired by the ideas of Ayn Rand, known for writing novels about individualism and laissez-faire capitalism, and built in an art deco style, the city immediately tells players something about the world's history and values (individualism, ambition, collapse...). The space reflects familiar concepts to the player and evokes a story.

Enacted narratives happen through the player's movement in the game world. Contrary to watching a cutscene or reading a story, players move through a designed path that aims at stimulating emotions like tension, and provide some kind of meaning. The narrative often appears through localised incidents or micronarratives that the player triggers by moving in the world. These kinds of stories are often more controlled by the game designers, but they work because they use the environment to

make players feel something while they play. For example, in *Fallout 4* (Bethesda, 2015), when the player approaches a specific location (Trinity tower), it triggers a sidequest that the player can choose to follow to rescue a character (Strong). During this mission, if the player chooses to rescue Strong (involving specific dialogue and actions where the player potentially helps the character). If the player successfully completes this quest, it results in Strong becoming a permanent companion, which influences the narrative by integrating him more deeply into the player's progress in the game.

Embedded narratives are pieces of story that are hidden in the environment. Players can find and piece them together if they explore closely. These fragments of stories might be objects, notes, audio logs, or even the way rooms are arranged. The narrative is built from clues. This kind of storytelling rewards players who take their time, explore, and think about what they see. It's often used to add depth or background to the main story. In *Fallout: New Vegas* (Obsidian Entertainment, 2010) during "The House Always Wins" series of quests, the player can find clues about the personality and motives of the local overlord, Mister House. They are embedded within the environment of New Vegas, through terminals, holotapes, and object layouts and written notes. As a result, players can reconstruct aspects of his story and reveal the character's ambitions.

On the other hand, emergent narratives are not planned by the designers. They "emerge" from how players interact with the game systems. These homemade stories are created by the player through their choices, actions, and experiences within the game. The game provides tools and rules, but the player's story is unique. This approach often appears in sandbox or open-world games where players have freedom to explore, build, or experiment with the gameworld such as in games like *Minecraft* (Mojang, 2011). In this game, there's no set story. But players create their own stories by building, surviving, and exploring. As I have discussed in section 3.1.1, *Minecraft* players can exhibit independent behaviours. One player will build a fortress and decorate it. Another will travel across the game world to find resources or discover new villages. The way players use the digital space echoes the story they tell. Easily, we can link it with Yi-Fu Tuan's (2001) notion of space and place. Evocative spaces in games often become meaningful through repeatedly moving inside this space and realising its connection with our past experiences. Similar to Tuan's idea that "space becomes place" as players attach memory and significance to specific environments. Likewise, Kevin Lynch's (1960) concept of imageability is reflected in how embedded narratives rely on spatial legibility. A well-placed landmark aids orientation but also carries narrative meaning. In the same way Sleeping Beauty's castle is placed as a central landmark in the attraction park of Disneyland Paris.

4.4.2 Narrative Verticality

Together, these perspectives illustrate how digital games structure narrative through spatial design. This understanding of narrative will inform the following analysis of verticality as a narrative tool that impacts the player's involvement with the game. Verticality is an architectural language through which games express ideas, emotions, and progression. Vertical movement evokes symbolic and experiential meanings that contribute to how players understand and feel the game world. Similarly, vertical structures are used to convey different emotions (e.g. the sensation of being at the feet of an imposing castle). This section explores how verticality supports environmental storytelling and contributes to player presence by deepening narrative engagement. To better understand how verticality functions as a narrative device in digital games, this section applies Henry Jenkins' (2004) typology of environmental storytelling. As observed in the previous section, Jenkins identifies four narrative modes: evocative spaces, enacting stories, embedded narratives, and emergent narratives. Each describes how space in digital games can communicate or construct a story. This framework offers a useful lens through which to explore verticality as narratives. It reveals how elevation differences and actions linked to verticality become expressive structures that organise symbolic meaning, guide player experience, and shape the narrative architecture of a game world.

Each mode will be explored through verticality. First, evocative spaces consider how vertical structures such as towers, mountains, or ruins resonate with mythological, cultural, or emotional associations. Ascending or descending through these places can evoke metaphors of character progression, failure, or power. The work of Yi-Fu Tuan (2001) and Edward Casey (2002) is particularly relevant here, as both highlight how physical orientation and spatial features contribute to affective memory and place-making. Second, enacting stories are linked to the player's physical engagement with vertical movement (climbing, falling, jumping) as a form of uncovering the narrative thought by the designer. These actions are symbolic and expressive, as players face challenges or struggles and develop their skills through spatial progression. Gordon Calleja's (2011) narrative and kinesthetic involvement, as well as Dubbelman's (2016) notion of narrative game mechanics, help illustrate how movement and actions by the player build a story. Close to this perspective, emergent narratives will be considered through verticality as a sandbox for storytelling, where players generate personal stories by navigating, manipulating, or altering the vertical environment. Games that offer vertical freedom of movement empower players to create their own narrative arcs through experimentation and progressing vertically. This form of storytelling, shaped by the player's actions, emphasises verticality as an enhancing condition for narrative improvisation. Finally, embedded narratives are examined through the design of vertical environments as information spaces. Here, height and depth are used

to structure discovery, curiosity and stimulate exploration (hidden caverns in the mountains, stacked story elements in a puzzle, or symbolic stratification). These spaces guide the player's understanding of the world's history and the character's backstory and any change that occurs. Gibson's (2015) notion of perception through affordances and Tuan's (2001) emphasis on the transformation of space into place through experience supports how meaning is embedded and revealed through vertical layering.

Together, these four modes provide a comprehensive framework for demonstrating how verticality becomes a central contributor to narrative experience, further grounding its role in shaping player presence. This section thus marks the final link in my exploration of verticality as a significant element for presence, showing how stories are communicated in space and also through moving across its vertical structures.

1. Evocative Spaces

In his work, Jenkins (2004) describes evocative spaces as environments that draw meaning not solely from original narratives, but from the way they echo existing stories, cultural symbols, and conventions. These spaces are rich in resonance because they activate the player's cultural memory, allowing them to step into worlds they have already encountered in myths, literature, societal standards or other games. Verticality enhances the evocative power of these spaces by drawing on spatial metaphors and cultural associations through vertical structures and architecture. Vertical structures in digital games are often used to evoke themes of aspiration, revelation, or personal growth when players ascend, while descent commonly signals danger, reflection, or confrontation with hidden treasures. These directional metaphors attempt to draw on familiar stories that players might have encountered outside of the game. This forms a compelling space that players intuitively understand. Tuan (2001) notes that space becomes a place through emotional investment and repeated engagement. In vertical game environments, this investment is intensified as players physically enact these metaphors, turning spatial traversal into a symbolic experience.

An example of this is *Journey* (Thatgamecompany, 2012), where a distant mountain stands as the central visual and thematic anchor. From the game's outset, the player's relationship with this peak is both spatial and emotional. The journey toward it becomes a meditation on effort and personal progression. The mountain is not explained with scripts, yet its vertical dominance makes it an immediately understood symbol of purpose, finality and even spirituality. With the player located in a desert and this dominating mountain in the distance, the developers seek to evoke a sense of smallness and wonder. The ascension towards the top becomes the story, supported by visual design,

sound, and physical progression. This symbolic function of verticality aligns with Mircea Eliade's (1958) concept of the Axis Mundi (a vertical connection between the earthly and the divine). In religious and mythic traditions, mountains, towers, and trees often function as sacred structures that bridge the human and the spiritual, further supporting that vertical structures play a significant symbolic role evoking stories.

This way, verticality also reinforces genre associations. In fantasy and adventure games, castles with towers, sky islands, or ruins are often places with quests or hidden rewards potentially guarded. For instance, the design of a gothic castle, historically used to impress and generate unease to establish dominance, carries almost immediate expectations based on its height and silhouette. Designers will expect players to anticipate some kind of mystery, seclusion, and engagement with the unknown. In the horror game *Until Dawn* (Supermassive Games, 2015), descending into the mountain cavern (home to the antagonist) or the house's cellars intensifies dread and vulnerability. All in one, it shows that verticality strengthens evocative space by enhancing its symbolic and cultural dimensions with layouts that resonate familiarity to players.

2. Enacting Stories

In his typology, Jenkins (2004) describes enacted narratives as those that emerge through the player's movement and choices based on authored stories performed through gameplay. Verticality directly supports this type of storytelling because the affordances it provides transforms movement through space into narrative expression. Climbing, falling, gliding or jumping are player-driven actions playing as instruments to progress in the game through success and failure. With these player decisions and physical acts, stories unfold. This can link to Calleja's (2011) kinesthetic and narrative involvement. As I have demonstrated earlier in this chapter, vertical gameplay intensifies kinesthetic involvement, requiring players to coordinate timing, momentum, and spatial judgment. However, these movements also stimulate the environmental narrative by bodily interacting with it. For instance a game designer can structure a climb to reflect a specific narrative arc that uncover a new quest, which merges with the player's personal experience that becomes an integral part of the story they enact. Verticality allows designed structures to suggest certain themes, while giving players the freedom to perform and embody the narrative of the game.

Moreover, verticality affects the resistance in the environment and the pacing it induces which further enhance enacted narratives. A location filled with enemies placed at different elevations, or wind currents used for navigating this space demands awareness and focus from the player, which in turns,

generates tension and suspense. These sensations are foundations of traditional narrative structures that aim for rising action and climax. Yi-Fu Tuan's (2001) notion is also relevant here. When players have a prolonged engagement with the virtual space, they become familiar with it and generate meanings linked to it. The space becomes a site of memory and narrative because of what the player enacted within it (meaning created through moving and acting). This shows that verticality supports enacted narratives by giving players opportunities to perform and embody story arcs through spatial navigation. Designers can construct vertical environments that suggest themes that allow the player's bodily interaction to animate them, make them speak. Stories can be climbed, jumped, and fallen through.

3. Embedded Narratives

Jenkins (2004) describes embedded narratives as pre-authored narrative elements placed within the environment, waiting to be discovered and interpreted by the player. These narrative artefacts, such as notes, architectural clues, objects, or decorations, rely on the player's spatial participation to be pieced together. Vertical environments offer a particularly effective structure for embedding narrative elements because they naturally organise information through layers, and hierarchy in space.

As seen in evocative spaces, vertical spaces in games often signal a movement from one narrative layer to another. Ascending can reveal new secrets, insight, or confrontation, while descending can uncover treasures, forgotten history or memories. This aspect can be experienced in *Mad Max* (Avalanche Studios, 2015) where artefacts of the past are often hidden in buried locations that feel like time capsules. This way, designers use verticality to structure the flow of narrative information, with secrets placed off the main path, high above or far below the expected route., which encourages the player to engage actively with the environment. For instance, *Control* (Remedy Entertainment, 2019) uses the vertical layout of the Oldest House as a maze but also as an archive of narrative information. Key documents, environmental details, and visual cues are often located in inaccessible upper floors, obscure corners, or basements that must be reached through spatial puzzle-solving or with new abilities that reveal the bureaucratic history, metaphysical anomalies, and surreal culture of the institution. It sets the narrative world. This structure draws on Gibson's (2015) theory, where the environment offers clues about possible actions. Vertical spaces in games suggest routes, risks and rewards that come with narrative implications. A locked door that seems to be unlockable poses questions on the history of this situation, and what could be hidden behind.

Lynch's (1960) theory of imageability further supports this approach. He emphasised that memorable environments are composed of legible features (landmarks, paths, nodes) that aid orientation and create meaning. Vertical structures naturally stand out as landmarks. Towers and cliffs guide movement and symbolise significance based on their architecture. The embedding of story within vertical environments also ties into Lefebvre's (1991) spatial triad. Conceived space corresponds to the designer's deliberate placement of narrative clues. Perceived space relates to how the player navigates and interacts with those placements. Lived space emerges when players interpret the significance of what they find. For example, in *Hollow Knight* (Team Cherry, 2017), descending into the ancient city of Hallownest reveals narrative elements like a decaying city with a rich history. Each cavernous level tells part of the story through environmental layering such as symbols on the walls, ruins of forgotten kingdoms, and fragments of dialogue heard from ghosts. Hence, this demonstrates that verticality is an essential structural design for conveying embedded narratives to the players, through architecture, different types of elevations and the feelings that are made from towering elements and the hidden secrets they afford.

4. Emergent Narratives

While I examined how verticality allows players to enact stories embedded in level design and spatial structure, it is important to acknowledge how closely this overlaps with emergent narrative. Both perspectives emphasise player agency and the embodied experience of moving, and both rely on movement through space to convey meaning. However, they differ in the degree of authorial control and the source of narrative meaning. Enacting stories illustrates the designer's intentional narrative, with players fulfilling or interpreting pre-established arcs through their actions. Emergent narrative, by contrast, foregrounds the improvisational behaviour from players, where meaning appears less from scripted elements and more from how players engage with the system, tools, and opportunities in the environment. In the case of verticality, both approaches are leveraged with vertical movements, but whereas one channels it through curated spatial metaphors, the other provides it to the player as a narrative sandbox where they write their own stories.

Vertical game environments introduce interesting elements that affect the way a player progresses: platforms, cliffs, towers, flight, detours, and improvisational problem-solving. Players must often determine how to climb a cliff, cross a canyon or reach a high vantage point, and in doing so, they create stories based on their imagination and the strategy they chose to put in action. These narratives are entirely unique to their playthroughs since others might have approached the same objective differently.

ToTK exemplifies this with its vertical systems (gliding, climbing, constructing flying devices). They allow for open-ended exploration and player expression. There is no single route to a location in this large open-world with so many different levels of height. Instead, players can create their own trajectories by evaluating the different vertical structures like mountains and towers and building flying machines to avoid these obstacles. For example, one player might construct a gliding machine to reach a floating island, while another might climb and use a Recall ability to ascend via a falling rock. These actions are ingenious and experimental, but they indeed participate in the story for progressing through the game. This emergent approach aligns closely with Calleja's (2011) spatial and ludic involvement. The player, when solving a puzzle or following a path, is exercising meaningful choices about how and where to move, and their physical journey becomes the narrative. Moreover, this ties back to Tuan's (2001) argument that space becomes place through repeated interaction and emotional investment. When a player revisits a mountaintop they once struggled to reach, or when sliding downhill became an escape route that allowed the player to survive. These actions and events created places, rich with memory, tension, and meaning that in turn generated a narrative merging with the space.

Emergent vertical narratives also benefit from the affordance structure of vertical spaces, which is supported by Gibson (2015). Zip-lines, vents, and liftable objects offer immediate clues about what is possible, while leaving the specifics of use and timing to the player. For example, in *The Finals* (Embark Studios, 2023), vertical movement via grappling hooks, elevators, and destructible structures allows players to invent unexpected paths to objectives. Taking down a whole building can be the quickest and most surprising method to steal an objective and get the win. Altogether, it shows that verticality in emergent narrative enables a form of personal storytelling through space. It is open-ended, performative, and unique to each playthrough. With verticality, players construct meaningful memories and actions which generate experiences that feel entirely their own. Verticality, therefore, functions as a significant element for emergent narratives.

This section has demonstrated the capabilities of verticality to act as a meaningful narrative vector for digital games. It is a medium that influences the creation of narrative through environmental storytelling based on the four dimensions described by Jenkins (2004) and supported by the introduced theories. For a practical overview of the application of this perspective, here is an extensive case study of *Fallout 4* (Bethesda, 2015) that will support the role of verticality as a significant narrative device.

Focus: Fallout 4

Bethesda's *Fallout 4* (2015) offers a useful example of how verticality contributes to narrative engagement through environmental storytelling. Using Jenkins' (2004) framework of environmental storytelling, the game demonstrates how vertical movement and architecture can convey narrative meaning. It also helps players interpret the environmental symbols and invite player-authored experiences.

Fallout 4 is a single-player open-world role-playing game developed by Bethesda and released in 2015. Set in a post-apocalyptic version of Boston and its surrounding areas, the game is part of the long-running *Fallout* series, known for its blend of retro-futuristic aesthetic inspired from the 50s and branching stories based on player choices. Players assume the role of the "Sole Survivor", who emerges from an underground vault to explore the remnants of society following a global nuclear war and are led by the determination to find their kidnapped son.

The game is characterised by its wide world, first- or third-person perspective, flexible and numerous quests, and a combination of combat, stealth, and dialogue-based interactions. It includes systems for base-building, crafting, and character progression via perks and SPECIAL attributes that affect the avatar's ability in the game world. *Fallout 4* does not have cutscenes (apart from the introductory one) and places significant emphasis on exploration and environmental storytelling, encouraging players to uncover the game's history through ruined structures, scattered terminals and notes, and space presented in different layers of elevation. Its vertical space design (e.g. skyscrapers, mines, underground vaults and bunkers) play a central role in how players engage with the world.

Evocative

Spaces

A good example of verticality functioning as an evocative space in *Fallout 4* is Fort Hagen, a heavily fortified underground military complex encountered during the main quest. This location is used as a setting for combat but it also taps into Cold War-era imagery of bunkers and hidden military installations, which evokes secrecy, power, and fear. Players have different choices to enter the place depending on which level they want to penetrate the building (rooftop, underground parking). Fort Hagen is a space that evokes both historical and fictional traditions of underground strongholds. It echoes ideas of surveillance, state control, and buried truth with plenty of military equipment and documents scattered at the different storeys of the fort.

The descent into Fort Hagen's basement feels like diving into the unknown and readying for confrontation at each corner (which happens as the place is guarded). The player must navigate

hallways and lower levels, sometimes poorly lit, which increases claustrophobia. Moreover, The deeper the player goes, the more secure and sterile the space becomes. This marks the difference with the chaotic ruins of the empty surface world. From this contrast emerges a sense of intrusion and danger, which reinforces the authority of the hidden forces the player is up against. Eventually the player confronts the antagonist, Kellogg. This downward movement is a figurative descent into the protagonist's personal history, as the player progresses in the basement, Kellogg speaks to them through speakers, also, Kellogg holds key knowledge about the missing son. This supports Yi-Fu Tuan's (2001) notion as the player experiences emotional investment and uncover meanings in reaching the objective deep in the basement. The next time the player goes to Fort Hagen, what they lived during this quest (memories about Kellogg and what was learnt) will be mirrored by this place.

Another example of verticality's evocative power in *Fallout 4* is The Prydwen, the Brotherhood of Steel's (one of the many factions in the game) iconic airship that looms above the Commonwealth skyline. As a structure suspended in the air, it evokes both technological superiority and ideological dominance in a desolate wasteland where pre-war technology has been lost and survival dominates everyone. In order to board the airship, players must ascend to it via a type of helicopter only available to the faction and allies, which reinforces its symbolic inaccessibility and authority. Once aboard, the rigid militaristic layout, metallic interiors, and command room used as a vantage point over the wasteland reflect the Brotherhood's hierarchical worldview. As with vertical structures described by Eliade's *axis mundi* (1959), The Prydwen functions as a floating seat of power that connects the struggle of the ground people with elevated ideology. This turns its position in the sky into a narrative symbol of control and aspiration or even fear and disgust as narrated by some non-playable characters (NPCs).

Drawing on Jenkins's (2004) framework, *Fallout 4* exemplifies how verticality can be used in evocative spaces to build narrative resonance. Verticality support spaces that are built with familiar topics like military secrecy, underground facilities, authoritarian control, social hierarchy which are translated into an interactive, explorable structure.

Enacted

Stories

Verticality in *Fallout 4* also supports narrative engagement by enabling players to enact meaningful experiences through movement and decision-making. The story unfolds through how the player navigates them. A clear example of this can be found in the Mass Fusion Building, a skyscraper in central Boston that becomes an important part of a late-game mission involving high-stakes conflicts between the different factions. The building's architecture presents a vertical challenge: multiple floors filled with enemies, broken staircases, environmental hazards, and blocked paths that push the

player to find alternative passages or solutions (e.g. finding and starting the generator to be able to use the main elevator) which also generate tension because the story surprises the player's expected narrative. Depending on the player's chosen faction the objectives and story framing change. The environment reacts to the player's allegiance, which offers different enemy placements, narrative dialogue, and consequences. But the structure remains the same: a multi-level space where the climbing, fighting and exploring effects the story's progression.

This kind of mission featuring vertical structure extends the ways of performing narrative. For example, the ascent through the tower becomes a way for the player to live the tension of the story. Additionally, the pace at which the player progresses in the tower, the choice of entrance (e.g. rooftop or entrance), the use of stealth or force, and the response to environmental obstacles, such as enemies dropping from the high ceiling or firing from vantage points, all shape how the narrative is experienced. Friendly fire is also an important aspect to keep in mind as attacking an ally will result in them becoming hostile, this is even more of a threat with the complexity of the vertical environment which brings many entities close to each other on a line of fire. Here, we can apply Calleja's (2011) kinesthetic and narrative involvement. When the player moves inside the space, they are engaged physically, by the challenge of navigation and combat, and narratively, because their actions carry consequences and symbolic weight.

The narrative experience through vertical movement is also reinforced by the atmosphere of the space. Collapsed walls reveal a glimpse of the city landscape, which gives a perspective of elevation. The tower ascension is accentuated by scattered audio logs, and high architecture that hints at the pre-war world and the facility's importance. When advancing through the location, players are piecing together the building's history, its collapse, and the ideological conflict it embodies. This makes an emotional experience to which players can relate. Altogether, verticality in *Fallout 4* allows players to enact and embody the story through their movement, decisions, and the conflicts they face. This makes the story personal and meaningful, which at term supports the overall narrative involvement of the player.



The elevator battle when entering the Mass Fusion building in *Fallout 4*

Embedded

Narratives

In *Fallout 4*, verticality is also a crucial narrative layer embedded within the environment. As Jenkins (2004) defines, embedded narratives refer to story elements that are built into the world and must be discovered through exploration and interpretation. These narratives are often spatially distributed and require the player to piece together meaning by observing environmental details, reading logs, or traversing carefully designed locations. Vertical environments in *Fallout 4* frequently act as vessels for these narratives, they structure how and when information is revealed as the player moves upward or downward.

Let us take back the example of the Mass Fusion building. The building is part of a major storyline, but even outside of the quest path, its vertical design reveals fragments of the world's history. Similar to what I put forward with enacted stories, when players ascend the structure (whether by elevator, staircases or collapsed ceilings) they encounter destroyed offices, shattered glass walls, and broken executive meeting rooms. Scattered terminals contain emails and memos that hint at internal disputes, ethical debates over fusion technology, and environmental cues showing the fear just before the bombs dropped. The benefit the vertical structure of the building is that each level adds more context and depth to the explorative experience, which enriches the central theme often recurring in the game of corporate pride and rogue scientific ambition.

Embedded narratives often act together with the other forms of environmental storytelling to create a compelling story that the player can interpret and enjoy. As Jenkins (2004, p.7) notes: “We may have to battle our way past antagonists, navigate through mazes, or figure out how to pick locks in order to move through the narratively impregnated mise-en-scene. Such a mixture of enacted and embedded narrative elements can allow for a balance between the flexibility of interactivity and the coherence of a pre-authored narrative”.

This vertical progression plays as a narrative arc that gives information on the game world. The higher the player climbs, the more they learn about Mass Fusion's role in the pre-war world. This culminates when the player accesses the executive suite or rooftop with more secrets to discover and hidden technology. The layout leads the player to interpret the building itself as a symbol of societal collapse. It was built on ambition, now in ruins and deserted because of pride and egocentrism. Jenkins (2004) suggests that well-designed spatial narratives condition the player's movement and interpretation, and this is clearly demonstrated through the building's spatial structure. This approach also resonates with Casey's (1997) idea that memory and story are not just told in time but situated in space. The Mass Fusion building functions like an archive. It stores fragments of the past (text, imagery, statues, etc.) that are unlocked by progressing through the tower. Ascending reveals the story, floor by floor, and thus the building becomes a medium of storytelling in itself. All of this demonstrates how verticality acts as a significant design tool to ameliorate embedded narratives.

Emergent

Narratives

Fallout 4 also provides rich opportunities for emergent storytelling through its open-world design. Emergent narratives arise from player-driven decisions, unscripted interactions, and environmental exploration. As Jenkins (2004) explains, emergent narratives are “not pre-structured or pre-authored” but generated through the player's movement, choice, and improvisation within a spatial framework. In this regard, verticality in Fallout 4 serves as a sandbox in which players create their own stories through interacting with the vertical space and the objects that it contains.

The Skyscrapers in Downtown Boston, such as the Custom House Tower or Hub 360, exemplify how verticality allows for player-authored experiences. These buildings are not always linked to quests, however, they invite exploration through their visual prominence and implied possibilities of reward such as rare weapons. Players can choose to investigate them out of curiosity, often discovering hidden loot at the culminating point of the building after a long struggle to fight its way through all the enemies at each level. The player probably will encounter small narrative details as they progress in the skyscraper embedded in environmental cues, such as skeletons having a specific pose which helps the player make their own imagination of what could have happened at this specific location.

An example in *Fallout 4* of this that uses verticality to enrich the narrative capacity behind the cues are two skeletons in pre-war clothes sitting on a chair next to each other on the rooftop, they are facing towards the horizon. By choosing to go on that rooftop, the player is rewarded with elements in the environment that stimulate their imagination and could make their own story of the place, also, the simple act of going there is constructing a story to the player where the found treasures and the setting of the environment participate in the player's imagination. Building on this, climbing to the top of these spaces often requires piecing together a path through collapsed floors, elevator shafts, fire escapes, and precarious walkways. The path is not always obvious, which encourages experimentation and nonlinear exploration. The player could choose to walk and climb up the tower, or could choose to install a jetpack on their armour and fly to the top without traversing the obstacles. This also contributes to the experience of the player which is different from each other. The story they made inside the game is subjective. This freedom supports Calleja (2011)'s spatial involvement as well as Tuan's (2001) and Casey's (2002) ideas. The sense of being embedded in and actively navigating a meaningful world which is influenced by tall, multi-level spaces. Players map their own routes, confront unexpected enemies, or use the vertical layout tactically, such as sniping from rooftops or setting traps in staircases. Another example of this can be found in player interactions with the Vertibird system. As the player progresses in the Brotherhood of Steel faction, they gain access to Vertibirds, fictional helicopters that allow for air-based deployment across the map. Players can use them to land on building rooftops, infiltrate from above, use the support of the aircraft's armament to fight, or bypass dangerous zones entirely. These availability of vertical entry points provide more strategic choices from which emerge unique playthroughs and player-based narrative developments.

The base-building system in *Fallout 4* is also a key aspect for providing emergent narratives to the player. This system gives a flexible, user-driven environment where players can create, customise, and manage settlements to craft their own stories. The way settlements are built echoes the narrative that players had in mind and want to illustrate. Furthermore, by incorporating verticality, players can design settlements that mimic realistic social hierarchies (the player's house can be more dominant compared to settlers'), defensible positions, or specialised zones (residential, industrial, commercial...). For example, a player can build guard towers on elevated foundations to symbolise a more militarised zone, or create rooftop gardens that tell a story about how the survivors are resourceful and adapted to the wasteland. Also, vertical spaces generate memorable moments such as defending a settlement attack from one of these crafted guard towers. These scenarios become personalised stories driven by player design choices and in-game events.

To conclude this case study, *Fallout 4* shows how verticality enhances narrative involvement by acting as a medium for space to tell stories. With verticality, players engage with narrative content through movement, which makes the experience more personal and embodied. Vertical spaces in the game embed story elements, encourage emergent play, and symbolically frame encounters, which enriches both emotional and interpretive engagement. In this way, verticality strengthens narrative involvement, which, as shown in previous chapters, plays a crucial role in fostering player presence.

4.5 The Verticality Effect

Throughout this chapter, I have shown that verticality in game design is far more than a mechanical or aesthetic element. It is a force that shapes player involvement across perceptual, kinesthetic, narrative, and emotional domains. Verticality enables players to act with space (climbing, gliding, falling, jumping, etc.) as physical tasks that create meaningful experiences. These actions demand awareness that in turn generate rhythm in the gameplay, construct memories for the player, and are a strong support for decision-making. In doing so, verticality deepens a player's sense of being in the world. This intensity, embodiment and personal experience that emerges from verticality foster player presence.

By investigating verticality through the lens of presence, I argue that its strength lies in how it activates multiple forms of involvement simultaneously. It connects the body, the mind, and the imagination. When a player climbs a tower in *Assassin's Creed II* (Ubisoft, 2009), use ziplines to access rooftops in *The Finals* (Embark Studios, 2023), or descends into the dark tunnels in *Mad Max* (Avalanche Studios, 2015), they are traversing game space that causes stories to be staged, and an environment that speaks to the player. Verticality merges the cognitive with the emotional.

But more than this, verticality in digital games speaks to a fundamental human impulse: to reach higher, to fall into the unknown, to seek new perspectives through movement in space. In real life, we scale mountains, climb trees, explore caves, and build cities that always reach higher altitudes more than for survival, but because these acts hold stimulate our senses. Standing at a summit gives us a sense of perspective, not just over terrain, but over ourselves. The view from above invites contemplation, awe, and often, comfort because we feel so small. After all, there is an adjective that expresses our lack of words for this sensation: "a jaw-dropping view". The climb itself becomes a story to tell with moments of effort, fear, and success. We take pictures from high places because the view is beautiful, but also because we want to remember how it felt to get there.

Digital games afford access to these experiences in meaningful ways, and verticality is an important vector. They allow us to perform the journey, to feel the struggle and the reward, to simulate risk and achieve fulfilment in worlds both impossible and familiar. Just with these words we identify the psychological ascend from the low grounds to the illuminating summit. Verticality, in digital games, satisfies this deep-rooted human desire since it grants players the ability to rise, fall, and discover, in spaces in which design and actions are lived.

What excites me most about this research is that it reveals how something as seemingly simple as “going up or down” or “the tall and the small” in a game can echo so much meaning. It shows that our behaviour in a digital game is not neutral or empty of sense. It is shaped by our cultural literacy, our intuitions, and our human desire. In that sense, verticality becomes a tool for understanding how we relate to space, to narrative, and to ourselves. If presence in digital games is about feeling truly there, then verticality is a powerful contributor to that sensation. It connects us with the space we move through and the stories we experience. And ultimately, it lets us “climb” toward a deeper understanding of why space, motion, and perspective matter in human experience.

Chapter 5: Conclusion

This thesis addressed the question of how verticality influences player experience and presence in modern 3D digital games. Through a multidisciplinary synthesis of theories on presence, space, and perception, I established a theoretical framework for analysing verticality as a significant design element. Chapter 2 laid the conceptual groundwork by navigating the academic discourse on presence. It culminated in the adoption of Calleja's (2011) player involvement model as the central analytical framework, providing a structured methodology to analyse how different modes of engagement are influenced by verticality. Chapter 3 then introduces the theories of space as a supporting element to link presence with verticality, drawing on key theories to understand its function in games. It explored how space becomes a meaningful place (Tuan, 2001), how environments are perceived through actionable affordances (Gibson, 2015), and how space is socially and culturally produced (Lefebvre, 1991). This established a vocabulary for discussing game worlds as lived, and interpreted environments. Finally, the core analysis in Chapter 4 applied these frameworks to verticality. It argued that vertical design shapes player experience by expanding affordances for movement and strategy (as seen in *The Legend of Zelda: Tears of the Kingdom* (Nintendo, 2023)), enhancing game feel and kinesthetic involvement (demonstrated with *The Finals* (Embark Studios, 2023)), and acting as a powerful narrative device. Using Jenkins' (2004) typology, the analysis of *Fallout 4* (Bethesda, 2015) showed how vertical traversal conveys meaning and structures narrative. Collectively, this demonstrated that verticality intensifies multiple dimensions of player involvement, thus making a significant contribution to presence.

Hence, the findings of this thesis converge on a central argument: verticality in modern 3D digital games is a multidimensional design layer that illuminates societal symbols, produces emotions and contributes to a more impactful experience, therefore acting as a catalyst for player presence. For this, I demonstrated that verticality enhances spatial and narrative engagement. The act of ascending or descending constructs meaningful places, creating landmarks within the player's personal experience, which is amplified by the capabilities of verticality to act in environmental storytelling. Verticality also enhances player agency by expanding affordances given to the player as it offers numerous "action possibilities" in the game world. This encourages problem-solving and enhances the player's sensation of being in control. Also, as shown, this agency is physical because verticality strengthens embodied interaction through a rich kinesthetic feedback loop. The physical sensations of vertical traversal create a heightened bodily connection with the avatar to which the player can more intuitively relate to. Ultimately, these intersecting effects create a synergy that culminates in a deeper sense of

presence. Verticality activates multiple dimensions of player involvement simultaneously, which consequently deepens this feeling of “being there” in the virtual world.

Implications

This thesis contributes to the academic discourse in game studies in several key areas. Primarily, it addresses a significant gap in the literature by providing a focused, in-depth investigation of verticality. While spatiality and presence are well-established topics in the field of game studies, the specific experiential and narrative implications of verticality have remained largely underexplored. By isolating and examining verticality, this research expands the academic conversation around space and presence, offering a more nuanced understanding of how level design with verticality in mind directly impacts player engagement. This research’s main theoretical contribution lies in its interdisciplinary synthesis. It systematically brings together frameworks from humanistic geography, psychology, sociology, architecture, and game studies to build a comprehensive foundation for analysing verticality. This demonstrates how a holistic approach can yield rich insights into the complex relationship between players and game worlds, which provides a transferable model for future research into other specific aspects of spatial design. Furthermore, with this thesis I offer practical insights into the tangible effects of vertical design on gameplay and understanding player experience. By connecting theoretical concepts to concrete case studies, I provide a vocabulary and analytical framework for designers and critics to discuss how verticality can be leveraged to enhance affordances, guide emotional responses, and deepen narrative integration in digital games.

Beyond its academic contributions, this research offers several practical implications for game designers and developers. First, when designing vertical traversal mechanics, developing a strong sense of kinesthetic feedback and game feel is crucial. The physical act of climbing, jumping, or gliding should feel responsive, weighty, and intuitive. As demonstrated, a well-tuned vertical movement system enhances player agency and strengthens the connection to the avatar, forming a crucial foundation for presence. Second, designed vertical spaces are useful to resonate symbolical and emotional associations. The choice to make a location towering and imposing or deep and claustrophobic is also a narrative and affective choice. By leveraging cultural associations with height and depth designers can infuse environments with meaning that players understand intuitively. This means considering the architectural language of vertical structures and how they guide the player’s journey in experiencing the game. Finally, verticality can be harnessed to deepen both narrative and gameplay dynamics. For narrative, vertical layers can be used to structure the flow of information, embed story elements in hard-to-reach places to reward exploration. For gameplay, verticality

introduces strategic depth, brings combat possibilities, complex environmental puzzles, and emergent player-driven solutions. By designing worlds with verticality, games can create dynamic sandboxes that encourage player-based gameplay and replayable experiences. Ultimately, a deliberate and thoughtful approach to verticality can transform game worlds into compelling, memorable, and engaging places.

Eventually, the significance of verticality in digital games taps into a fundamental aspect of human experience. Humanity has long been drawn to height by climbing mountains and erecting towers to create new perspectives, face challenges or get closer to our spiritual beliefs. This vertical drive finds its roots into our cultural traditions that symbolises ambition, the need for exploration, and the confrontation with our own limits. Digital games, as this thesis has shown, are powerful mediums for simulating this drive. They provide accessible and repeatable grounds to experience the core sensations of verticality such as the risk of a climb and vertigo, the exhilaration of flight, and the discovery of something new. Virtual verticality is therefore a meaningful way for players to engage with powerful, archetypal experiences. Therefore, the vertical spaces are sites where players connect with a broader human desire to relate to the world.

Limitations and Future Research

It is important to acknowledge the limitations that define the scope of this study. This research has been primarily theoretical and qualitative, focusing on building a conceptual framework and applying it through interpretive analysis of selected case studies. Its focus on modern 3D games, while necessary for a detailed examination of complex spatial navigation, means its conclusions may not be directly transferable to 2D, isometric, or other game genres where verticality functions differently. The use of a small number of case studies, though allowing for in-depth analysis, also means the findings are illustrative rather than empirically generalisable across the entire landscape of digital games.

These limitations, however, open up several promising avenues for future research. There is a clear need for quantitative and empirical studies to test the claims made in this thesis. Researchers could design experiments to measure the correlation between specific vertical mechanics and biometric indicators of presence, such as heart rate or galvanic skin response. Comparative studies would also be valuable, examining how verticality is implemented and experienced across different genres (e.g., platformers vs. open-world RPGs) or between different cultural contexts in game design. Finally, the emergence of virtual and augmented reality (VR/AR) presents a particularly fertile ground for future investigation. Exploring how the fully embodied perception of height and depth in VR environments

impacts player presence, comfort, and kinesthetic engagement would be a logical and exciting extension of the work started here.

Closing Thoughts

My personal motivation for this research was born from my own experience playing games like *The Legend of Zelda: Breath of the Wild* (Nintendo, 2017) or *Just Cause 3* (Square Enix, 2015). The ability to climb any surface or soar through the air, this bodily and mental engagement to deal with what the vertical space provides us, this feeling of being able to go anywhere. This led to a simple curiosity: how can something as fundamental as moving up or down in a game echo with so much meaning? This thesis has been an attempt to answer that question, revealing that our interactions within digital spaces are shaped by the same intuitions, cultures, and desires that guide our experience of the real world. The power of verticality lies in this very connection. Vertical spaces are resonant sites of feeling and being. When we climb, fall, and fly in a virtual world, we are engaging in a practice that connects us to a long history of human exploration and aspiration. In these moments, digital games posit their capacity to be more than just entertainment as they become spaces where we can experience, in a tangible and embodied way, what it means to reach for something higher.

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